



Lettre de motivation – Candidature au poste de professeur en sociologie quantitative – Université Laval

Madame, Monsieur,

Je vous soumetts ma candidature au poste de professeur en sociologie et méthodologie quantitative au Département de sociologie de l'Université Laval. Je suis actuellement Senior Institutional Researcher (poste permanent) à l'Université de la Colombie-Britannique (UBC), au sein de l'unité de planification et d'analytique institutionnelle. J'ai obtenu mon doctorat en sociologie à UBC, suivi d'un stage postdoctoral à Sciences Po, Paris au Centre de Sociologie des Organisations (CSO-CNRS-Institut d'Études Politiques de Paris). Ma formation est profondément ancrée en statistique sociale, en analyse quantitative, et science des données enrichie par un parcours collégial antérieur en sciences appliquées et fondamentales au Collège Jean-de-Brébeuf, ainsi qu'une formation universitaire en génie électrique et logiciel à l'Université de Sherbrooke. Je travaille couramment en français, en anglais et en mandarin.

L'ensemble de mes contributions porte sur la production des savoirs, les marchés académiques du travail et les inégalités scientifiques, en mobilisant principalement des méthodes quantitatives, complétées par des approches mixtes et des outils de science des données. Mes travaux ont donné lieu à 4 articles évalués par les pairs, 3 prépublications (preprints) et 2 chapitres d'ouvrages collectifs. Mon article le plus récent, « Working groups, gender, and publication impact of Canada's ecology and evolution faculty » (Wei, Lachapelle, Fuller & Srivastava, 2025), publié dans FACETS, analyse l'impact des structures collaboratives et des dynamiques de genre sur la production scientifique à l'aide de méthodes quantitatives avancées, incluant des modèles de survie de type Cox (proportional hazards) et des régressions à effets fixes avec splines linéaires. Un autre de mes articles, « Replacing the Canadianization generation: An examination of faculty composition from 1977 through 2017 » (Lachapelle & Burnett, 2018), publié dans Canadian Review of Sociology / Revue canadienne de sociologie, a reçu le Best Student Paper Award de l'Association canadienne de sociologie. Mes travaux ont par ailleurs été soutenus par plus de vingt bourses et prix de recherche, incluant des financements majeurs du Conseil de recherches en sciences humaines du Canada (CRSH/SSHRC) et du Fonds de recherche du Québec – Société et culture (FRQSC/FQRSC).

Ma thèse de doctorat examine l'émergence de nouvelles formes de production et de validation des connaissances scientifiques dans le contexte de la pandémie de COVID-19, en mettant l'accent sur les serveurs de prépublications. À l'aide de méthodes statistiques avancées, incluant la régression logistique et des algorithmes de jumelage à grande échelle, j'ai montré comment ces dispositifs reconfigurent les hiérarchies épistémiques traditionnelles et transforment les mécanismes d'accumulation du capital scientifique. La thèse met également en évidence des inégalités genrées dans l'accès à la production et à la reconnaissance scientifiques, contribuant ainsi aux débats contemporains sur la sociologie des sciences et du travail académique.

À la suite de mon doctorat, j'ai effectué un postdoctorat au Centre de sociologie des organisations de Sciences Po, dans le cadre d'un projet international dirigé par Emmanuel Lazega portant sur l'institutionnalisation transnationale de la Cour unifiée du brevet (Unified Patent Court). Ce projet mobilisait l'analyse de réseaux sociaux, la sociologie économique et la géographie judiciaire afin



d'étudier la structuration relationnelle des élites juridiques européennes. Cette recherche a donné lieu à une publication collective analysant la géographie judiciaire des litiges en matière de brevets et les processus d'harmonisation institutionnelle à l'échelle européenne.

Mes projets de recherche actuels s'inscrivent dans la continuité de ces travaux. Le premier, intitulé « Don't Publish, Don't Perish », examine les logiques atypiques de production des connaissances en économie académique, en combinant analyses statistiques longitudinales et entretiens qualitatifs. Un second projet majeur, Subfield, prolonge un article primé publié en 2017 et vise à constituer la plus vaste base de données recensant l'ensemble des professeures et professeurs universitaires au Canada. Ce projet mobilise des méthodes avancées de science des données et de modélisation statistique afin d'étudier les processus de recrutement, de prestige académique, de mobilité et de diversité dans l'enseignement supérieur.

Parallèlement à mes activités de recherche, j'occupe actuellement un poste stratégique à l'Université de la Colombie-Britannique, où je mène des analyses quantitatives à grande échelle portant sur les parcours étudiants, l'évaluation institutionnelle et les politiques universitaires. Mes travaux incluent notamment des analyses de représentativité des enquêtes, mobilisant des tests statistiques et des mesures d'association (dont le V de Cramér) afin d'identifier et de corriger les biais de non-réponse. Je développe également un manuscrit comparatif analysant les écarts méthodologiques et substantiels entre le suivi des trajectoires professionnelles des diplômé-e-s à partir d'instruments traditionnels (enquêtes) et de sources numériques alternatives, telles que les profils LinkedIn. Cette expérience renforce mon engagement envers une sociologie quantitative rigoureuse, appliquée à des enjeux concrets de gouvernance, d'équité et de production des données dans l'enseignement supérieur.

J'ai également acquis une solide expérience en enseignement et en encadrement. J'ai co-enseigné un cours de maîtrise en analyse de réseaux sociaux à Sciences Po, donné des cours et conférences en méthodes quantitatives et supervisé des projets de fin d'études en science des données. J'ai encadré des étudiantes et étudiants aux cycles supérieurs dans des contextes interdisciplinaires, en mettant l'accent sur la rigueur méthodologique, la reproductibilité et la formation statistique.

Enfin, je tiens à souligner mon attachement profond au Québec. Originaire de Sainte-Foy et ayant grandi entre la région de Québec et Brossard, je souhaite vivement contribuer au rayonnement académique de l'Université Laval et à la formation des étudiantes et étudiants québécois en sociologie quantitative.

Je vous remercie de l'attention portée à ma candidature et je me tiens à votre disposition pour toute information complémentaire.

Veuillez agréer, Madame, Monsieur, l'expression de mes salutations distinguées.

François Lachapelle, PhD

A handwritten signature in cursive script that reads 'François Lachapelle'.

François Lachapelle

Curriculum Vitae

Citizenship: Canadian born in Ste-Foy, QC city.

Languages: French, English, Mandarin, Python

Email: francois.lachapelle@ubc.ca

CURRENT EMPLOYMENT

2023-2026 Senior Institutional Researcher {full-time & permanent position}
Department of Planning, Analytics & Institutional Research.PAIR | Predictive Analytics Unit
University of British Columbia – Vancouver Campus, BC, Canada

RECENT POSITIONS

2021-2023 Post-Doctoral Fellowship
Institut d'Études Politiques de Paris {SciencesPo}, Paris, France
Centre de Sociologie des Organisations (CSO-CNRS)
Project: Emergent Transnational Institutionalization of the European Patent Court – UPC
Fields: SNA, Economic Sociology, Work and Professions, Sociology of Elites, STS
Professor: Dr. Emmanuel Lazega

2021-2023 Post-Doctoral Researcher
University of British Columbia – Vancouver Campus
Dept. of Sociology
Project: U.S. NSF – “Can We Change Organizational Overwork Culture”
Fields: Economic Sociology, Work and Professions
Professors: Dr. Elizabeth Hirsh & Dr. Youngjoo Cha (**Indiana University**)

EDUCATION

2014-2022 Ph.D., **University of British Columbia**, Dept. of Sociology, BC, Canada
Dissertation: *Scientific Knowledge Production and Academic Labor In Unsettled Times : COVID-19, Epistemic Validation, Preprint Servers, and Gendered Work*
Supervisor: Dr. Elizabeth Hirsh
Committee: Dr. David Tindall, Dr. John McLevey, and Dr. Rima Wilkes
Comprehensive Examination Areas: Work; Culture & Theory

2011-2014 M.A., **University of British Columbia**, Dept. of Sociology, BC, Canada
Thesis: *From Nameless Marxist to Public Sociology: The Intellectual Fate of Shen Yuan in Post-Mao China*
Supervisor: Dr. Amy Hanser
Committee: Dr. Timotyh Cheek and Dr. Neil Gross

2005-2008 B.A., **Université de Montréal**, History and East Asian Studies (Honor), Montreal, Canada

2001-2004 B.Eng., **Université de Sherbrooke**, Electrical/Software Engineering. (32 courses completed)

1998-2000 D.E.C. {College Degree}, **Collège Jean-de-Brébeuf**, Applied and Pure Science (CÉGEP), Montreal, Canada

SUBSTANTIVE RESEARCH INTEREST

- Sociology of knowledge and science
- Inequalities in science
- High-status labor market, professions & work
- Higher education
- Post-Mao intellectual history

QUANTITATIVE/METHODOLOGICAL APPROACHES

- Social Statistics (regression, survival models)
- Social Network Analysis (SNA) & ERGMs
- Supervised Machine Learning (ML)
- Computational Social Science / Natural Language Processing (NLP)
- Small & Large Language Models (LLMs)

PEER-REVIEWED ARTICLES

Wei, Qian, **François Lachapelle**, Sylvia Fuller, & Diane S Srivastava. (2025). "[Working groups, gender, and publication impact of Canada's ecology and evolution faculty](#)". *FACETS*, 10:1-11.

Zipf, Marius, Johannes Gucker, Tamar Khuchua, Emmanuel Lazega, **François Lachapelle**, & Jakob Hoffmann. (2023). "[The judicial geography of patent litigation in Germany : Harmonizing regional variation in legal practice](#)". *Social Sciences*, 12(5), 311.

❖ Article selected as the Issue Cover for the Month of June 2023.

Lachapelle, François and Patrick J. Burnett. (2018). "[Replacing the Canadianization generation: An examination of faculty composition from 1977 through 2017](#)". *Canadian Review of Sociology/Revue Canadienne de Sociologie*, 55(1): 40-66.

❖ *Best Overall Student Paper Award*, Canadian Sociological Association, 2017.

Shi Anshu, **François Lachapelle**, and Matthew Galway. (2018). "[The recasting of Chinese socialism: The Chinese New Left since 2000](#)". *China Information*, 32(1): 139-159.

❖ First written in Chinese language and then translated in English, summer 2015.

Lachapelle, François. (2017). "Shen Yuan, public sociology, and nameless Marxism in post-Mao China." *Chinese Perspective*. Special Feature on the Chinese Social Sciences versus Western Social Sciences debate: an epistemological account.

❖ Accepted for publication in August 2017, but Special Issue was cancelled.

DATA DEPOSITION

Wei, Qian, **François Lachapelle**, Sylvia Fuller, & Diane S Srivastava. (2025). "[Data from Working groups, gender, and publication impact of Canada's ecology and evolution faculty](#)". Borealis.

PRE-PRINT ARTICLES

François Lachapelle. (2022). "[Media Securitizer and the Racist "China-Threat" Discourse: Ascension, Peak, and Downfall](#)". *SocArXiv*, 1-55.

François Lachapelle. (2020). “COVID-19 Preprints and Their Publishing Rate: An Improved Method”. *medRxiv*, 1-34. DOI: <https://doi.org/10.1101/2020.09.04.20188771>.

- cited 21 times {as of Nov 2025}

Qian, Yue, **François Lachapelle**, Diane S. Srivastava, Sylvia Fuller, and Catherine Corrigall-Brown. (2020). “Working groups, gender, and publication impact of Canada’s ecology and evolution faculty”.

BioRxiv, xx-xx. DOI: <https://doi.org/10.1101/2020.05.12.092247>.

- cited 2 times {as of Nov 2023}

BOOK CHAPTERS

Lazega, Emmanuel & **Lachapelle, François.** (2023). Chapter 2 : Collegial Oligarchy and Democratic Deficit in Emergent Transnational Institution-Building – The Case of European Patent Judges Assembled at the Venice Forum. In Fernard de Visscher, Alain Strowel, Vincent Cassiers, & Luc Desautnettes (eds) *The Future of the Unified Patent Court in Europe*. Ledezioni Publisher.

Lachapelle, François. (2021). Chapter 8: COVID-19 and Science. In K. Lyon & K. Christhy (eds.). *COVID 19 & Society*. Oxford University Press.

MANUSCRIPT(S) IN PROGRESS

Lachapelle, François. “Don’t Publish, Don’t Perish, or The Organized Laziness of Academic Economists”. *Target Venue: Social Forces*.

Lachapelle, François. “Judicial Reputation, Litigation Experience & the Spectre of the Unified Patent Court: The Case of France First Instance Court – 2008 to 2021”. *Target Venue: The European Journal of Sociology*.

OTHER ACADEMIC PUBLICATIONS

Lachapelle, François and X. Alvin Yang. (2025). “[Status hunt drives U.S. recruitment](#)”. *University Affairs*. June 2025.

Lachapelle, François and Patrick J. Burnett. (2018). “Canadian Universities: Between domesticity and globality”. *Global Dialogue* 8(1): 16. Translated in 15 languages.

- ❖ Via national competition.

Lachapelle, François. (2016). “The Genesis of a Chinese public sociologist”. *Global Dialogue* 6(2): 23-24. Translated in 15 languages.

- ❖ Invitation to contribute by Micheal Burawoy.

ACADEMIC AND POLICY REPORTS

Qian, Yue, **François Lachapelle**, Diane S. Srivastava, Sylvia Fuller, and Catherine Corrigall-Brown. (2020). “Working groups, gender, and publication impact of Canada’s ecology and evolution faculty”. Grant-Report for the National Sciences and Engineering Research Council of Canada. 17 pages.

Lachapelle, François. 2015. “AMS Report on Canadian Professoriate - The Canadianization Movement Revisited: Canadian Professoriate, Envy-League and the Social Sciences Between 1978 and 2008, ” for Pierre Cenerelli, UBC AMS University & Government Relations Advisor.

TEACHING EXPERIENCE

Winter Term 1 – 2023: Introduction to Social Network Analysis (Masters’s I level), *SciencesPo-Paris*, Centre de Sociologie des Organisations

- Co-instructor with Dr. Emmanuel Lazega

Winter Term 1 – 2022: COVID-19 & Society Course, UBC, Dept. of Sociology

- Instructor responsible for Science, Knowledge, Culture, & Society modules
- Instructional Materials published as:
 - **Lachapelle, François.** (2021). Chapter 8: COVID-19 and Science. In K. Lyon & K. Christhy (eds.). *COVID-19 & Society*. Oxford University Press.

MENTORING EXPERIENCE

Summer Term – 2025: UBC-Okanagan, Dept. of Computer Science, Mathematics, Physics, & Statistics

- Supervisor and mentor for 2 Master of Data Science (MDS) capstone teams (8 students), providing guidance on project design, machine learning, data engineering architecture, documentation, methods, and statistical analyses.

2021-2023 : *SciencesPo-Paris*, Centre de Sociologie des Organisations

- Supervised and mentored 1 undergraduate & 1 Masters-level research assistants in data collection/standardization, and social network analysis.

2018-2019 : UBC, Dept. of Sociology

- Mentored 2 undergraduate international & Indigenous students in navigating post-graduation career choices, graduate school applications, and research methodology.

TUTORIAL EXPERIENCE

Introduction to Python Tutorial I & II in Dr. Oral Robinson’s Research Methods, UBC, Nov. 2019.

- ❖ First Python Tutorial organized by a graduate student for undergrad sociology students in sociology dept. (UBC).

Introduction to SPSS & Statistics [4 tutorials] in Dr. Sylvia Bertolic’s Survey Research Methods, UBC, Oct. 2018.

TEACHING ASSISTANTSHIP EXPERIENCE

UBC, Department of Sociology:

SOCI217 – Research Methods, Dr. Oral Robinson, 2016 – 2019

- 15% of Teaching Responsibility: Big Data, Measurement, Experiment.
- Organization of first Python tutorial series in home department, 2019.

SOCI317 – Survey Research Methods, Dr. Sylvia Bertolic, 2018

- 15% of Teaching Responsibility: SPSS and Statistics Analysis.

SOCI350 – Classical Sociological Theory, Various Instructors, 2013 – 2016.
 SOCI320 – Canadian Society, Instructor Dr. Rima Wilkes, 2014 – 2016.
 SOCI100 – Introduction to Sociology, Instructor Dr. Stéphanie Skoutes, 2014.
 SOCI217 – Introduction to Research Method, Instructor Dr. Sean Lauer, 2013.
 SOCI250 – Crime and Society, Instructor Dr. Carole Blackburn, 2011.
 SOCI342 – Culture and Consumption, Instructor Dr. Carris Yodanis, 2011.

GUEST LECTURES

Guest Lecturer “*What is Big Data?*” in Dr. Oral Robinson’s Research Methods, UBC, May 2017, Nov. 2019.

Guest Lecturer “*Conceptualization and Operationalization: the Case of the Canadianization Movement,*” in Dr. Oral Robinson’s Research Methods, UBC, April 2017, Sep. 2019.

Guest Lecturer “*The Messiness of Being in the Field: Doing Fieldwork in Your Field in Contemporary China,*” in Dr. Dawn Currie’s Feminist Research Methods and Theory Course, UBC, December 2012.

PROFESSIONAL HONORS, AWARDS, & FELLOWSHIPS

Scholarly Work [Post-Doctoral Level]:

- Science Po Paris ANR-DFG Post-Doctoral Fellowship, 2021-2023.

Graduate Work:

- UBC President’s Academic Excellence Initiative PhD Award, 2021 & 2022.
- UBC Sociology Dissertation Writing Scholarship (\$6,000), 2019/2020.
- Patricia Marchak International Research Scholarship in Sociology (\$880), 2019 & 2020.
- UBC Faculty of Arts Graduate Award (\$2,497), 2019.
- Kurt and Anne Paulus Memorial Scholarship in Sociology (\$500), 2019.
- Best Student Paper Award, Canadian Sociological Association (\$250), 2016.
- UBC Public Scholar Initiative (\$8,000), 2016-2017.
 - Support doctoral students whose dissertation push the boundaries of traditional scholarship.
- UBC Four Year Doctoral Fellowship, 2014-2018.
- AMS Impact Grant, UBC (\$3,000), 2014.
- FQRSC (Quebec Research Funds for Society and Culture) Bourse de Maîtrise en Recherche B1 (\$15,000), 2012-2013.
- UBC Institute of Asian Research Student Fellowship, First Cohort, (\$1,500), 2013.
- SSHRC Joseph-Armand Bombardier Canada Graduate Fellowships Program Master’s Fellowship (\$17,000), 2011-2012.
- UBC Faculty of Arts Graduate Award (\$3,500), 2012.
- SSHRC Micheal-Smith Foreign Study Supplements (\$6,000), 2012.
- UBC Faculty of Arts Graduate Award (\$10,000), 2011.
- Canada-China Scholars’ Exchange Program Fellowship (\$18,000), 2009-2010.
- TECO (Taiwan Economic and Cultural Office) in Canada Ministry of Education Huayu Enrichment Fellowship Funding Offer (\$15,000), 2009-2010 & 2015-2016. [Declined twice]
- Quebec-China Fellowship (\$18,000), 2008-2009.
- Quebec Government Mobility Fellowship (\$5,000), 2006.
- East Canadian Oral Debate In Chinese Competition, McGill - Second Place, 2006.

Undergraduate Studies:

- PIE-FAEUM (\$800), 2007.

- Prix Forces Udm – Catégorie Science et Technology (\$1,800), 2007.
 - Won for the organization of two video-conferences series.
- Dean’s List for Outstanding Academic Records, Udm, 2006-2008.

LANGUAGE TRAINING

Full-Time Chinese Language Student:

Summer 2012 Berkeley IUP (Inter-University Program), **Tsinghua University**, Beijing, People’s Republic of China (PRC)

2009-2010 **Sichuan University**, Chengdu, Sichuan, PRC (full academic year)

2008-2009 **Harbin Institute of Technology**, Harbin, Heilongjiang, PRC (full academic year)

Summer 2006 **Nankai University**, Tianjin, PRC

Modern Manchu Language:

2010-2012 UBC, Dr. Carla Nappi, History Department

LANGUAGES SKILLS

Human languages: French (native); English; Chinese (mandarin; advanced / cantonese; beg.); Manchu (beg.)

Computer languages: Python [advanced], C [interm.], R [interm.], Bash [beg.], Julia [beg.], Perl [beg.], Lisp [beg.], 6502 Assembly [beg.]

PROFESSIONAL PRESENTATIONS

“Peeking Under the Hook of the Data Science Labor Market”. June 2025 & October 2025. University of British Columbia, Depart. of Computer Science, Mathematics, Physics, and Statistics., Master of Data Science Program Professional Development Series.

“UBC Students – Birdeye View”. May 2024. University of British Columbia – Okanagan. Celebrate Learning Week.

“*Doctoral Origins and Academic Labor in the Periphery of American Sociology*”. June 2023. CSA Session on Indigenizing, decolonizing, or globalizing sociology in a multi-polar world.

“*A Relational Look at the Work of High-Status Judicial Actors: The Case of French Patent Actors*”. June 2023. CSA Session on Professions and Occupations: Inequalities and Complexities.

“*Media Securitizer and the Racist "China-Threat" Discourse -- Ascension, Peak, and Downfall*”. Virtual Event. May 2022. CSA Session on Race and Ethnicity.

“*Predicting COVID-19 Preprint Manuscript's Publication Status: Scientific Impact, Social Attention, Gender, and Institutional Prestige*”, Virtual Event. May 2022. CSA Session on Sociology of Knowledge.

“*Collegial Elite in Emerging Transnational Institution-Building. The Case of European Intellectual Property Judges Assembled at the Venice Forum*”, Brussels {Belgium}, January 28, 2022. The Unified Patent Court System: Remaining Weaknesses and Possible Adjustments Conference. Session on Legality and Legitimacy Failures.

“COVID-19 preprints and the Future of Scientific Labor”, Alberta, June 2021. CSA Section on Work and Occupation.

“The Gender Gap in COVID-19 Research Authorship”, Alberta, June 2021. CSA Section on Feminism and Patriarchy.

“The Rise and Stall of the Canadianization Movement: Inequalities among Canadian Professoriate, Evidences from 1977-2017,” Montreal, August 2017. ASA Section on Science, Knowledge and Technology Refereed Roundtables.

“From Nameless Marxist to Public Sociology: The Liminal Trajectory of Yuan Shen in Contemporary China”. Seattle, August 2016. ASA Public Sociology Panel.

“Evidence of a de-Canadianization in Selected U15 Social Science Departments.” Toronto, May 31, 2017. CSA Invited Session Panelist: Challenges to the integrity of academic hiring practices in the corporate university.

“Tracing the Doctoral Origins of PhDs Who Makes it Into the Professoriate Class in Canada: Empirical Evidence from 1977-2017”, with Patrick Burnett. Toronto, May 29, 2017. CSA Roundtable - Issues in the Sociology of Education.

“The Canadianization Movement Revisited: Canadian Professoriate, Envy-League, and the Social Sciences”. Calgary, June 2016. CSA Work and Occupation Panel.

“From Nameless Marxist to Public Sociologist: The Liminal Trajectory of Yuan Shen in Post-Mao China”. Calgary, June 2016. CSA Sociology of Knowledge Panel.

Homo Academicus: ‘Who gets the job ?’ Engaging Education for Public Good. PhDs Go Public Event. UBC Learning Commons, Vancouver DES. January 26th, 2017. <https://www.youtube.com/watch?v=18zT4VTY8xl> (Starts at 1:02:37).

The Rise and Stall of the Canadianization Movement: Canadian Professoriate, Envy-League, and the Social Sciences, Evidences from 1977-2017. UBC Higher Education Research Group, Octobre 21st, 2016.

The Global Canadian Academia Beyond Methodological Nationalism: Exploring Burris’ Relational Hypothesis of Networks of PhDs Exchange. Science & Academic Careers : Summer School in Higher Education and Science Studies & German Center for Higher Education Research and Science Studies (DZHW). Hanover University, (Germany), September 5-9, 2016.

“Sino Academicus: Towards Mapping Fields of Intellectual Politics in Contemporary China,” at Book Project Working Group ‘Reading and Writing the Chinese Dream’, UBC, Institute of Asian Studies, August 15-22 2015.

“Avant l’Héritage et la Critique: l’École de Qinghua et la Sociologie Pratique de la Civilisation Communiste,” at (Re)faire la Sociologie: Entre Héritage et Critique, Colloque des Cycles Supérieurs de Sociologie, Montréal, March 25, 2011.

CONFERENCES ORGANIZATION

Growing Pains: Challenges to Development in Northeast Asia, Video-Conference Series with UBC, UfT, McGill, and Udm, Organized by François Lachapelle, David Passerilli, and CETASE Director, David Ownby, Udm, 2008.

In the Shadow of a Monolith: Contemplating the Rise of Asia's New Giants, Video-Conference Series with UBC, UFT, McGill, UDM, Waseda and Harvard, Organized by François Lachapelle, David Passerilli, and CETASE Director, David Ownby, UDM, 2007.

RESEARCH ASSISTANTSHIP EXPERIENCE

RA-1: UBC, Dr. Julian Dierkes, Institute of Asian Research, 2012

- Univariate and Bivariate analyses (Demographic and Survey Questions).
- Report of the State of Japanese Studies in North America Report for the Japan Foundation.

RA-2: UBC, Dr. Amy Hanser, Sociology, 2013

- Lit. Review for Article Project, 2013.

ACADEMIC AND ASSOCIATIVE LIFE SERVICES

1. **Co-Technical Lead**, British Columbia Student Outcomes Survey Forum Management Team, 2023—now
2. **Data Officer**, UBC Graduate Student Council Society, 2018—2021
 - Founding officer. Creating a longitudinal database tracking the career outcome of all UBC sociology MA and PhD programs graduates between 1998 and 2018.
3. **Senior Associate Editor**, Sojourners: Undergraduate Journal of Sociology, 2014—2015
 - Helping department's undergrad journal editors selecting manuscripts for review and publication.
4. **Graduate Mentor**, UBC Graduate Student Council of Sociology (GSCS) Mentorship Program, 2014—2015.
 - Mentoring First-Nation students with graduate school applications.
5. **Editor-in-Chief**, Le Cétasiatique, Université de Montréal CÉTASE Students Newspaper, 2007-2008.
6. **Vice-President**, ASSO-CÉTASE, Université de Montréal East-Asian Student Association, 2006-2008.

REVIEWER & COMMUNITY SERVICE

ROR (Research Organizations Registry) Community Advisory Committee
Scientometrics
Social Networks

PROFESSIONAL AFFILIATIONS

Canadian Sociological Association
American Sociological Association
UBC Higher Education Research Group
UBC Big Data and Computational Social Science Research Cluster
UBC Conceptual History Reading Group
UBC Chinese Language Translation Group
E@UBC

APPLICATION PROGRAMMING INTERFACE {API}

Upload-or-Publish Database (2020—ongoing)

- Free, Micro-API to query a COVID-19 preprints' publication status
- 16,000+ preprints' publication status available (last update: March 2021)
- The current beta-version allows one route → solo /DOI query:
 - http://heibufan.pythonanywhere.com/json/pp_meta/doi
- Documentation
 - <https://github.com/lanbufan/Upload-or-Publish>

DATASETS

Lachapelle, François (2020). Subfield.lab *Global Census of Scientific Workers*

- First wave of data collection underway
- First attempt to create a census dataset of academic workers using only publicly available data

Lachapelle, François. (2020). Subfield.lab *Daily Scientific Citations Tracker*: a Longitudinal dataset tracking the references/citations count of each Covid-19 related-publications with a daily updates.

- First daily tracker of scientific citations developed in the history of science

Lachapelle, François. (2020). Subfield.lab *Link-Cov-P Database*: a daily updated and comprehensive linkage of data sources on Covid-19 research with extensive meta-data information [13, 240 unique cases- as of 01.05.2020].

- Record Linkage of all major ongoing databases on Covid-19 research: CORD-19, LitCovid, WHO, Academia.edu, WOS

Lachapelle, François, Yue Qian, and Diane Stivastava. (2020). *SynSciSocio Publications Database* : a Longitudinal Dataset of Canadian Ecology and Evolution Publication, 1953-2020 [90, 100 unique cases].

Lachapelle, François, Yue Qian, and Diane Stivastava. (2020). *SynSciSocio Researchers Database* : a Longitudinal Dataset of Canadian Ecology and Evolution Researchers, 1953-2020 [1, 314 unique cases].

Lachapelle, François and Patrick J. Burnett. (2018.) *U15*: a Longitudinal Dataset of Canadian Professoriate's Educational Credentials, 1978-2017 [4, 365 unique cases].

- Won the 2017 CSA Best Student Paper
- Largest national dataset detailing the educational background of university professors.

PROFESSIONAL POSITIONS

subFIELD.lab

2013—Present

Founding Director and Lead Data Developer

- subFIELD is a higher-education start-up focuses on building datasets and ranking systems to help universities better benchmark themselves and keep universities more accountable.
- Leading the development of the first global census of university-based researcher.
- Supervising the conception of the first global ranking of university's PhD placement performance, gender, race, and institutional diversity.

Canadian Institute for Ecology and Evolution

May—September 2019

Data Science Consultant

- Building a complete publication records of 1, 400 Canadian biologists using several linked datasets and more than 1.2 million data-points for Web of Science.
- Using machine-learning techniques to predict which publication(s) can be categorized as synthesis science.
- Leading a team of 3 undergraduate research assistants and one post-doctoral fellow.

Science Outside the Lab North

Oct—Dec. 2018

Data Science Consultant

- Using Python code to build a Canadian census of every director of department in 55 universities.

C2C

Sept. 2016—Dec. 2017

Lead Researcher

- Funded by Mitacs Accelerate Grant, UBC, 2016.
- Leading the full research cycle—grant application, ethics, survey design & administration, write-up.
- Objectives: gaining insights into the early career trajectory of UBC 2015 graduate cohorts.

ACADEMIC BLOG ENTRIES

1. Lachapelle, François. 2009. “Rainy Days in Harbin.” Blogspot. CHINA AT OUR GATES : Blog of David Ownby, Professor of History and Director of the Center for East Asian Studies at the Université de Montréal. (<http://chinaatourgates.blogspot.ca/2009/06/rainy-days-in-harbin.html>).

French Version:

Lachapelle, François. 2009. “Réactions à Chaud : Temps Pluvieux sur Harbin.” (<http://lachineanotreporte.blogspot.ca/2009/06/reactions-chaud-temps-pluvieux-sur.html>).

2. Lachapelle, François. 2009. “From the Invisible Commemoration of Tiananmen to China’s Festive 60th Birthday.” Blogspot. CHINA AT OUR GATES. (<http://chinaatourgates.blogspot.ca/2009/06/from-invisible-commemoration-of.html>).

French Version:

Lachapelle, François. 2009. “D’une invisible 20e commémoration du 4 juin 1989 au festif et bruyant 60e anniversaire dynastique.” (<http://lachineanotreporte.blogspot.ca/2009/06/dune-invisible-20e-commemoration-du-4.html>).

3. Lachapelle, François. 2009. “Suivez le Guide – Tianjin (ou ne le suivez pas).” Blogspot. CHINA AT OUR GATES. (http://lachineanotreporte.blogspot.ca/2009/09/suivez-le-guidetianjin-ou-ne-le-suivez_14.html)

Working groups, gender, and publication impact of Canada's ecology and evolution faculty

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Abstract

Working groups are recognized as a highly effective method for synthesizing science. It is less clear if participating in working groups benefits individual researchers, or if benefits differ between men and women. This is a critical question, for the working group method is not sustainable if the benefit to science comes at a cost to academic careers or gender equity. Here we analyze the publications of Canadian university faculty specialized in ecology and evolution ($N = 1244$), a field that has embraced the working group method. Researchers were more likely to have participated in a working group as their academic age and prior H-index increased, but controlling for these factors there was no effect of gender. Using a longitudinal analysis, we find that researcher H-indices accrue 14% faster following their first working group publication, regardless of gender. Part of this acceleration may be the 3- to 5-fold higher citation rate of working group synthesis publications. In a survey ($N = 169$), researchers also report indirect benefits of working groups, at similar rates for men and women. Working groups are therefore good not just for science but also for scientists. Formalized mechanisms for collaborations such as working groups may also offset gender inequities in science.

Key words: synthesis science, working group, H-index, collaboration, gender equity, academic careers

Introduction

Progress in science depends on our ability to draw out general principles from large amounts of heterogeneous data. While the grist of science is empirical observations and experimental manipulations—that is, primary research—the full strength of individual studies is only realized when they are synthesized, either statistically, mathematically, or conceptually. Rapid development of computational tools to analyze large datasets, combined with the increasing availability of data through public repositories, has allowed scientists to harness the power of large-scale syntheses of previously published results (Hampton et al. 2013). Such “synthesis science” allows researchers to determine which processes are truly general (by comparing multiple studies) and to develop new paradigms (by exploring the interface between disciplines).

As synthesis science is often conducted through collaborative methods, the rise in synthesis science may also be driven by the increase in research conducted in collaborative networks. The ascent of collaborative research networks can be seen, for example, in the impact of the Human Genome Project or the Intergovernmental Panel on Climate Change, or more generally in the rise in the number of authors per publication (Wuchty et al. 2007; Huang 2015). Collaboration promotes research productivity and publications produced

by collaborative networks have a higher impact than those produced by individual researchers (Bordons et al. 1996; Lee and Bozeman 2005; Wuchty et al. 2007; Abramo et al. 2009; Gazni and Didegah 2011).

In the fields of ecology and evolution, synthesis science is often conducted through a specific type of collaboration research network known as working groups (WGs). These consist of a small network of researchers—typically 5 to 15 people—that meet to work intensively on a critical problem that requires the synthesis of large amounts of information and ideas and often involves insights from multiple (Rajan and Subramanian 2008) disciplines. Although WGs can be funded through a variety of mechanisms, most WGs in ecology and evolution are formally organized and funded through synthesis centers (Baron et al. 2017). The WG method for synthesis science has been highly influential in ecology and evolution. Specifically, the publications from WGs: (1) are cited more than other papers on similar topics; (2) bridge different knowledge clusters (subdisciplines) more often than other studies; and (3) play a pioneering role in the development of a new fields by synthesizing overall findings and resolving methodological issues (Halpern et al. 2020). Although science may benefit from the WG method, it is not clear whether individual scientists that participate in WGs also

benefit in terms of their academic careers. This matters, as the WG method is only viable in the long term if the interests of both science and scientists are aligned.

There are several reasons to expect that WGs may catalyze the careers of individual scientists by accelerating their publication rates and publication impacts. First, because WG publications are often highly influential (Halpern et al. 2020), participating in WGs may raise the publication impact of all participants. Here, the high impact of WG publications could either occur because synthetic research, the type of publication typically created by WGs, is cited more than primary research (Miranda and Garcia-Carpintero 2018), or because publications from collaborations are generally cited more than those produced individually (Lariviere et al. 2015; Leimu and Koricheva 2005). Second, there may be longer lasting impacts on careers as researchers develop new paradigms, construct large databases, and build collaborative networks that define and accelerate future research activities. Alternatively, participation in WGs may take time away from the primary research activities of researchers, depressing their overall publication rate.

Therefore, the first aim of this paper is to examine the impact of participation in WG on researchers' publication impact. As our analyses are based on observational data, it is difficult, if not impossible, to establish causality. If researchers who participate in WGs have stronger publication records than those who do not, it is not clear if this association reflects an effect of WGs on the careers of researchers or if stronger researchers are simply selected to participate in WGs. We can start to decouple this association by first testing for selection effects using the entire population of researchers who might feasibly participate in WGs, and then conducting a longitudinal analysis (publication record before vs. after WG participation) of just those researchers who have participated in WGs.

If there are career benefits emerging from WGs, another crucial question is whether access to these benefits is equitable. Patterns of collaborative research can disadvantage women, with some analyses suggesting they are less likely to be integrated into international research networks and to assume leadership roles in interdisciplinary research centers (Bozeman and Corley 2004; Misra et al. 2017). For example, when international collaboration requires travel, gendered family constraints may disproportionately prevent women's participation (Uhly et al. 2017). Furthermore, women tend to receive less recognition than their male collaborators despite similar contributions—a phenomenon known as the “Matilda Effect” (Rossiter 1993; Sarsons 2017). Collaboration can benefit science but not female scientists, and whether this is also true for WGs is far from clear.

Alternatively, women may receive more benefits from WGs organized via synthesis centers than less formalized types of collaboration. A number of synthesis centers have formal policies about gender balance for WGs, some going back to the 1990s. Some centers ask for diverse teams and use gender balance as an assessment criterion (e.g., National Center for Ecological Analysis and Synthesis, John Wesley Powell Center for Ecological Analysis and Synthesis) whereas other centers mandate minimum proportions of women and non-binary re-

searchers amongst the participant's team (e.g., Canadian Institute of Ecology and Evolution, German Centre for Integrative Biodiversity Research). The presence of formal policies should help address gender inequities in participation which may in turn help erode biases linking masculinity with perceived competence (Stainback and Tomaskovic-Devey 2009; Tomaskovic-Devey and Avent-Holt 2019). However, formal mandates do not always effectively counter interactional dynamics that can disadvantage marginalized groups. Subtle bias can persist, and even be magnified, where presumptions of meritocracy prevail, and mandated equity and diversity initiatives can prompt backlash (Castilla and Benard 2010; Dobbin and Kalev 2016; Kang et al. 2016). Women may participate in WGs at equal rates, but not in ways that ensure equitable credit via authorship in immediate research outputs. They may also fail to reap the same indirect career benefits of WG participation such as the development of useful network connections that facilitate future research and publication activities.

In what follows, we investigate the relationship between WG participation and career publication impact with a gendered lens. We ask: (1) Is the probability of participating in WGs affected by previous publication records or by gender? Do men and women accept/decline opportunities to participate at the same rate and for the same reasons? (2) Does participation in WGs lead to an increase in the publication impact of ecology and evolution researchers? If so, does this benefit differ for men and women? (3) Finally, do synthesis publications from WGs have a higher impact than other publications because of the type of science (synthesis vs. primary) or because of the method (WG vs. traditional), or both?

Methods

Data

Our study population is all Canadian university faculty members who received at least one NSERC Discovery grant through the Evolution and Ecology evaluation committee from 1991 to 2019 ($N = 1244$). NSERC Discovery Grants are the main source of research funding for ecology and evolution researchers in Canadian universities, providing a useful means of identifying active researchers in this area. We realize that some researchers who consider themselves ecologists or evolutionary biologists hold Discovery Grants through other committees. It is also possible to have a successful research career while never holding a Discovery Grant. Nonetheless, this is a simple and unambiguous criterion for defining our study population. We focus on this group of scientists for three reasons: (1) ecology and evolution researchers are strong users of the WG method. The first synthesis science center in ecology and evolution, NCEAS (National Center for Ecological Analysis and Synthesis) was established in the U.S. in 1995 (Hampton and Parker 2011) and the number of similar synthesis centers worldwide has now grown to 16; (2) Canadian researchers have been active in synthesis science WGs from the beginning, with WG publications appearing soon after the first synthesis center was established. Canadian researchers have participated in WGs in Canada and

internationally. Canada's own synthesis center, CIEE (Canadian Institute of Ecology and Evolution), started in 2008 and since then more than 500 researchers have taken part in WGs organized by CIEE; (3) women comprise 23% of this study population; i.e., there is a gender imbalance to examine, but still enough women for meaningful statistical analysis.

We collected information on researchers' assumed gender, year of PhD and all affiliated institutions (to aid in the matching of researcher names and outputs) through their careers from publicly available sources (Supplemental Information: Appendix 1). We coded gender (man, woman) from first names (where strongly associated with gender) and from pronouns and photographs on current websites. We acknowledge that such assumptions necessarily have a small degree of error: we may have mis-assigned gender in some cases or failed to capture non-binary or dynamic gender identities. The online sources for biographic information, from the most preferred to the least preferred, included curriculum vitae, websites of the researcher's current institution, personally-maintained website of the researcher, websites of the researcher's former institution(s), LinkedIn, Research Gate, Google Scholar, and other sources such as obituaries. Missing data are unavoidable since some information is not available from open sources. We initially identified 1405 researchers from the NSERC Discovery Grants database. However, 161 entries were excluded due to missing data, resulting in a final sample of 1244 individuals.

For each researcher, we quantified the trajectory of their publication impact using the H-index. A researcher has an H-index of H when they have published at least H publications, each of which is cited at least H times. The rate at which a researcher's H-index increases through time describes the growth of their career publication impact. Please note that our focus on H-index as a response variable in our analyses should not be taken as advocacy for this index as a reliable indicator of scientific impact. Instead, we use this index simply because it is so widely used—or misused—as an indicator of impact (Koltun and Hafner 2021). Indeed, our analyses add to previous work that shows systemic gender bias in this index.

We constructed each researcher's longitudinal H-index using (1) a retrospective publication record of their peer-reviewed articles and (2) the yearly distribution of citation counts for each publication. To our knowledge, there is no free, publicly available, and readily formatted data source that captures the H-index over time at the individual-researcher level.

To reconstruct a retrospective publication record of each researcher, we started with the Web of Science Core Collection (hereafter, WOS) as 36% of researchers, especially older researchers (modal PhD year = 1972), had no other online publication record. There are several challenges in developing this publication record. First, because WOS uses initials of all names but the surname, there is a possibility of erroneously including publications by other scholars with similar names (e.g., Jane Doe vs. John Doe), a type of false positive. Second, researchers with middle names can vary in which initials they include in their name across publications (e.g., a WOS search on JM Doe would not include publications under J Doe), a type of false negative. Finally, researchers may

change their names during their publication career (e.g., with marriage). We developed a Python-based workflow (described in full in Supplemental Information: Appendix 2), to reduce both the false positives and false negatives within the constraints of WOS. In brief, this workflow uses the researcher's full last name and first initial of their first name (based on all name variations in our database) to download from WOS all publications starting from 5 years prior to their PhD until 2019. This creates a large pool of potential publications (mean of 504 publications per researcher), which is subsequently filtered to a clean subset by cross-referencing with known variants in authorship names for the researcher (from online curriculum vitae or Google Scholar profile) as well as their institutional affiliations, fuzzy matching of publication titles from curriculum vitae or Google Scholar profile where possible, and recursive identification of previously unidentified affiliations to fine-tune the cross-referencing procedure. Once we had cleaned the publication record, we then calculated cumulative citations over years for each publication from WOS yearly citation counts as a precursor to calculating the H-index.

Our analysis also required us to identify which publications are products of WGs. To achieve this, we matched WOS titles with known WG publications funded by the 15 synthesis centers that comprise the International Synthesis Consortium (publication lists from synthesis center websites or obtained directly from centers), or by searching the funding and acknowledgement sections of publications for synthesis center names or acronyms (Supplemental Information: Appendix 3, Table A1). We captured WGs funded through other organizations or mechanisms by searching for keywords commonly used to describe WGs ("WG", "synthesis group", "synthesis WG", "synthesis committee", "synthesis workshop", "catalysis group"). All publications were then manually validated by two researchers experienced in biology and synthesis research, and coded as primary research versus synthesis research, and as WG method versus non-WG method. We further categorized synthesis research publications into the following types: statistical synthesis (statistical analysis of previously published or archived data collected by multiple different researchers and/or studies), conceptual synthesis (qualitative review of the literature or proposal of new frameworks for scientific concepts or investigation), or mathematical synthesis (theoretical mathematical models or specific application of general models for the purpose of prediction).

We scored non-WG publications using similar criteria. However, given the large number of publications involved, we changed methods to allow for programmatic approaches to identify publications based on keywords indicative of the three types of synthesis science (Supplemental Information: Table A2). This process yields 2541 new WOS titles that were then manually validated.

To provide additional contextual information on gender and WGs we conducted an online survey (Supplemental Information: Appendix 4) of ecology and evolution faculty in Canada using our researcher database as the sample frame. Of the 1244 ecology and evolution faculty members in our researcher database, we were able to find 1151 researchers' email addresses. An email invitation was sent to these

researchers containing a link to an online questionnaire with a consent cover letter. The survey was carried out from July to September 2019. Two rounds of reminders were sent to improve the response rate as well as in-person recruitment at the Canadian Society of Ecology and Evolution annual conference (Fredericton NB Canada, 18–21 August 2019). After clearing invalid questionnaires with too much missing data or no identification information, we had 169 valid responses, for an effective questionnaire response rate of 14.7%. The questionnaire (Supplemental Information: Appendix 4) asked for information designed to confirm or complete the researcher database (e.g., academic history, gender) as well as information about why researchers participated or not in WGs, and the perceived costs and benefits of participation.

Statistical models

The survey results were evaluated with simple chi-square tests of association.

The effects of research type (synthesis vs. primary) and method (WG vs. traditional) on publication citation rates were evaluated with zero-inflated generalized linear model based on a negative binomial error distribution with a log link (R package *glmmTMB*). The zero-inflated term was modelled as the logarithm of years prior to 2019 (year of data collection), as we reasoned that some recently published papers would not yet have had time to be cited even if they eventually will be. We excluded publications from 2019 or which drew equally from primary research and synthesis science approaches.

Our statistical analysis of the relationship between H-indices, gender, and WG participation was more complex. This analysis proceeded in two stages, a first stage that examines which researcher characteristics (prior H-index, gender) predict WG participation, and a second stage that tests for changes in researchers' H-index trajectory following participation in their first WG.

Survival analysis

Our first research question is whether gender or prior H-index predict which researchers participate in WGs. Of the 1244 researchers in our database, 183 (15%) had participated in at least one WG and obtained at least one identifiable publication from that WG. We used survival analysis to test if gender or pace of career progression (H-index adjusted for time since PhD) predicts the hazard rate of participation in WGs. Survival analysis is a statistical technique used to examine the expected time until an event of interest occurs, such as death, failure, recovery, or other specified events. In our study, the event is the scholar's first WG participation experience, with the rate representing how fast this participation occurs at a specific time. We included an interaction between gender and H-index to assess whether potential selection effects tied to research records captured by the H-index are the same for women and men. In our sample, the first WG publication was in 1993, so researchers enter the risk set either in the year 1993, or in the year of their PhD graduation for those who finished their PhD after 1993.

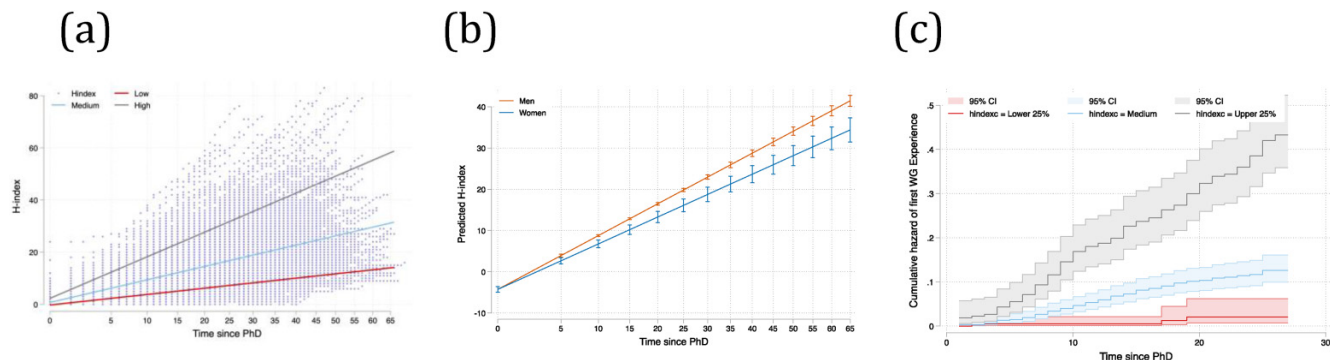
We estimated hazard ratios for attending WGs using Cox proportional hazards models. This model takes the form: $\log[h(t)] = a(t) + B'X$, where $a(t)$ is a function of time; hazard rate ($h(t)$) is the rate of participating in a WG, given that the researcher has not yet participated; X is a vector of covariates and B the vector of related coefficients. Since the left-hand side of the equation takes the logarithm of the hazard rate, we use hazard ratios ($\exp(B)$) as opposed to coefficients (B) to make our interpretation more intuitive. Specifically, hazard ratios represent the relative change in the rate associated with a one-unit change in a covariate, holding all other variables constant: a hazard ratio above one represents an increased rate, while a hazard ratio below one suggests a decreased rate.

Fixed effects regression with linear spline

Our second analysis focuses on the subset of researchers who participated in WGs. We employed a fixed effects model with a linear spline to investigate the effects of WG participation and gender on researchers' trajectory of H-indices over time. We chose a fixed-effects model since this technique nets out all unobserved time-invariant heterogeneity of individuals. In other words, each individual is treated as their own control in a longitudinal analysis (Allison 2009). In our case, this meant comparing the trajectory of researchers' H-indices in years before and after participating in WGs, and then averaging those differences across researchers. This approach ensures that we do not confound the effect of WG participation on researcher H-index with the effect of H-index on researcher selection for WG participation. To account for autocorrelation within individuals and heteroscedasticity across individuals, we clustered on individuals as suggested by Wooldridge (2016) to obtain robust standard errors. Although for most researchers, the increase in their H-indices over time appears almost linear, overall citation rates have increased dramatically in recent decades. If we factor out the effects of calendar year, H-index trajectories actually grow more slowly as time since PhD increases. We therefore implemented a 0.67 power transformation on the "time" variable to linearize the relationship (the 0.67 exponent was estimated from a log H-index versus log time model incorporating calendar year effects).

To model the temporal structure of H-index growth, we use a linear spline with three knots to allow for differing H-index growth trajectories in the years prior to and following researchers' first WG experience. Unlike a simple linear regression that assumes a constant slope over the entire period, a linear spline is a piecewise linear function that allows the slope to change at predefined time points, i.e., knots. In this context, the knots divide the time trajectory into three segments, enabling us to model differing growth rates of H-index before and after the first WG experience. Specifically, the first segment of time is 0–5 years before the first WG; the second segment is 1–5 years after the first WG; and the third interval is 6 and more years after the first WG. We code the spline specification in marginal form, which makes interpretation simple: coefficients of the second and third interval capture changes in H-index growth rates from their prior intervals.

Fig. 1. (a) The H-index of each individual researcher increases over time, but the rate of this increase varies substantially between different quartiles of H-index. (b) On average, the rate of H-index accrual over time is faster for men than women. Error bars are 95% confidence intervals on predictions. (c) The cumulative hazard that a researcher participates in their first working group (WG) increases with H-index. Coloured lines in (a) and (c) indicate quartiles of H-index accrual rates amongst researchers. In (a) and (b), the horizontal axis has a nonlinear (square root) scale to linearize the H-index accrual rates.



Results

The researcher population

We represent the career publication impact of researchers by their H-index, which summarizes the number of papers cited a minimum number of times. The H-index of researchers in our database varied substantially (Fig. 1a). We partitioned our data into quartiles based on the annual H-index from Year 0 to Year 66 after the PhD. While our sample includes a maximum of 69 years post-PhD, the years 67, 68, and 69 post-PhD with only two, one, and one observation, respectively, were excluded from our quartile calculations. The top 25% of researchers with the highest H-index scores were classified as “High H-index”, whereas the bottom 25% with the lowest scores were labeled “Low H-index”. The remaining 50% in the interquartile range were designated as “Medium H-index”. The mean H-index in the upper quartile is 25 (i.e., at least 25 papers cited at least 25 times) whereas the lower quartile records a mean H-index of 5.

In our researcher population, gender and H-index progression are not independent: women have significantly slower H-index progression than men (Fig. 1b). Women are also underrepresented among the ecology and evolution faculty, as revealed both in our survey (33.1%) and researcher database (23.15%). The database likely has a lower proportion of women than the survey because it includes older cohorts of researchers (higher proportions of men) that are no longer active or deceased, whereas we recruited survey respondents by email, and younger cohorts of researchers are more likely to have functional and public emails.

Influence of H-index and gender on working group participation

We start our analysis by assessing whether previous H-index or researcher gender affects the chances that a researcher participates in a WG. In our researcher database, 2.9% of women and 11.8% of men have WG experience. However, we cannot directly compare these percentages, because of potentially confounding effects of gendered patterns in

Table 1. Results of a Cox model estimating the hazard of a researcher experiencing their first working group publication (log likelihood = −1209).

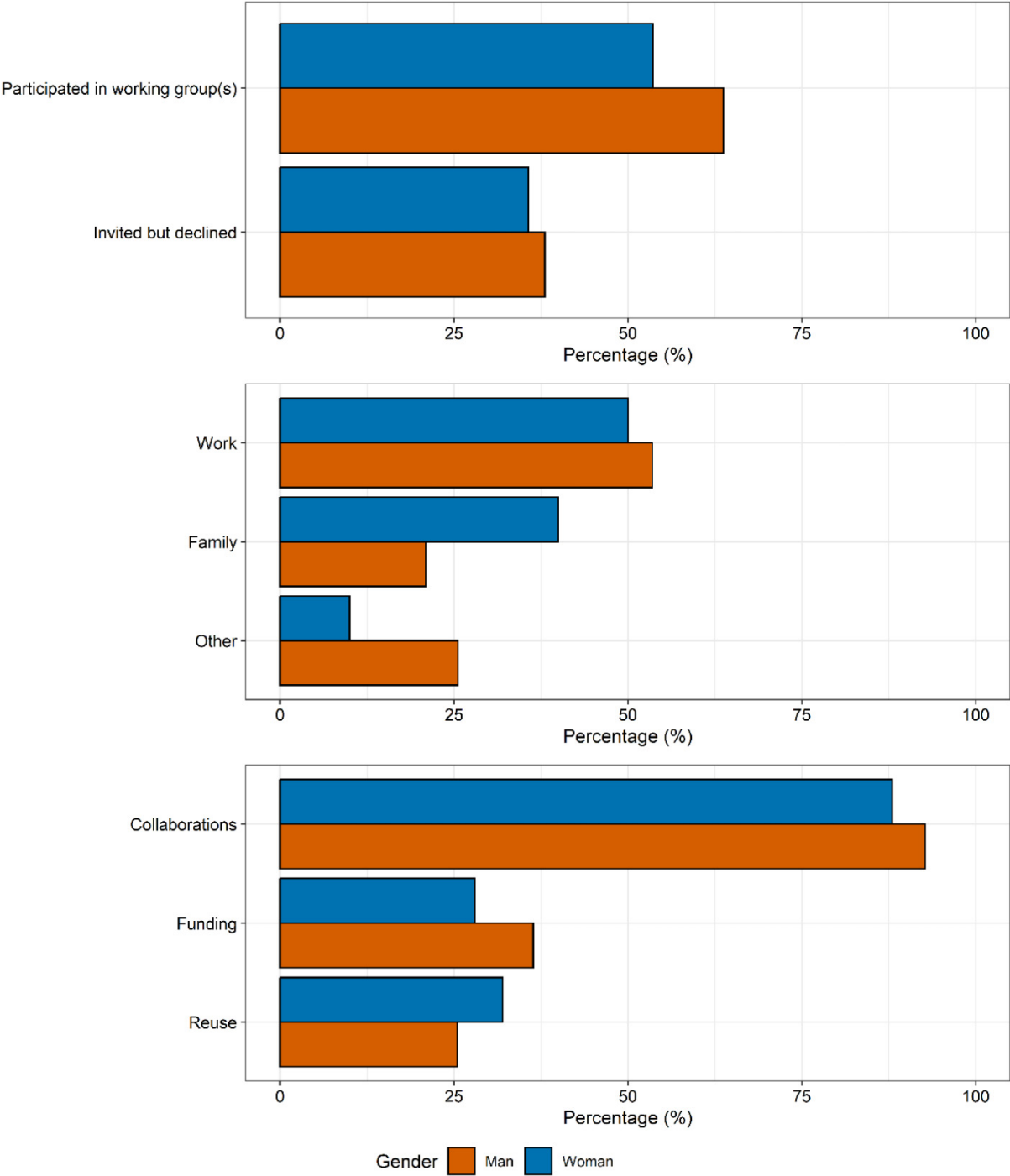
	Hazard ratio
H-index	1.07***
Woman	1.24
Woman*H-index	0.98
Time since PhD	0.93***

Note: Data are from the period 1993–2019 as no working group publications were found in our database in earlier years. Over this period, 183 out of 1244 researchers had at least one working group publication. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

researcher age (quantified here as time since PhD) and, as we have just shown, H-index. Instead, we conduct a survival analysis, the results of which indicate a strong association between higher H-indices and increased participation rates in WGs. We first applied the nonparametric Nelson–Aalen estimator and found a significant difference in the cumulative hazard rates of the initial WG experience across High, Medium, and Low H-index categories (Fig. 1c). The results of Cox model, listed in Table 1, demonstrate that the hazard ratio of the H-index is 1.07, which means that a one unit increase in H-index is associated with a 7% increase in the hazard rate of participating in the first WG. Meanwhile, neither gender nor its interaction with H-index are significant predictors of WG participation, suggesting there is no significant difference between men and women in their probability of participating in WGs once we account for academic age.

Findings drawn from our survey data echo this result: the men and women who responded to the survey have similar rates of participating in WGs. The majority of the 169 faculty who took part in our survey had participated in at least one WG (Fig. 2a): 54% of women researchers, and 64% of men researchers (a non-significant difference: $\chi^2_1 = 1.2$, $p = 0.27$). A similar proportion of women (36%) and men (38%) declined at least one invitation to take part in a WG

Fig. 2. A survey of 169 current Canadian faculty in ecology and evolution reveals that: (a) there is little gender difference in the percentage of faculty who have ever participated in a working group or have ever declined an invitation to participate; (b) of the 63 faculty who declined a working group invitation, the reasons given (work constraints, family constraints, other) are similar between men and women; (c) women and men reported similar rates of indirect benefits from working group participation, including subsequent collaboration with working group participants in other contexts (“collaborations”), funding opportunities (“funding”), and reuse of a database constructed within the working group for other projects (“database”).



($\chi^2_1 = 0.016$, $p = 0.90$). Both women and men listed “work-related duties” as the main reason for declining an invitation, but the second-most common reason for women researchers was “family-related duties” whereas men were at least as

likely to give other reasons, such as the research topic being outside their interests or expertise. Despite these trends, the reasons for declining WG participation do not significantly differ between men and women ($\chi^2_2 = 3.06$, $p = 0.22$)

Table 2. Longitudinal analysis (2564 observations) of H-index accrual in the 183 faculty with working group experience.

Spline regression with fixed effects of H-index on working group experience and gender		
	Model 1	Model 2
Years since first WG publication		
Pre-working group (−5, 0)	1.44***	1.52***
Post-working group (1, 5)	0.20*	0.22*
Long post-working group (6, 6+)	−0.13	−0.23
Woman* pre-working group (−5, 0)		−0.44***
Woman* post-working group (1, 5)		0.028
Woman* long post-working group (6, 6+)		0.33
Constant	11.56***	11.56***

Note: Spline regression model 1 ($r^2 = 0.86$) and model 2 ($r^2 = 0.87$) was used to test for changes in the H-index accrual rate, relative to that 5 years prior to the first working group publication (“pre-working group”), at specific points in time (years relative to first working group publication given in parentheses): immediately after the first working group (WG) publication (“post-working group”), and 5 years later (“long post-working group”). The effect of researcher gender being woman as opposed to man (“woman”) on H-index accrual rate, pre- and post- working group was also included in the model. Significance (alpha = 0.05) is demarcated as: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

Working groups and H-index progression

Since our survival analysis indicates that the subset of researchers with WG experience is biased with respect to prior H-index, analyzing H-index trajectories in the full population would confound effects of H-index on WG participation with effects of WG participation on H-index progression. Instead, we focus our analysis on longitudinal change in individual H-index trajectories within the subset of 183 researchers who participated in WGs. We use a time spline with knots at the time of WG participation, and 5 years later to evaluate the immediate and longer-term impacts of WG participation on the rate of H-index growth (Table 2). Model 1 reveals that WG experience is associated with a steepening of researchers’ H-index trajectory (Fig. 3a). Specifically, WG participation is associated with a 14% increase in the rate of H-index progression ($p = 0.034$); the H-index of a researcher with WG experience grows by 1.64 per year, while in the 5 years before participating, their H-index is predicted to increase by only 1.44 per year.

This acceleration in researchers’ H-indices was realized within 5 years of the first WG experience but does not change further after the 5 years (i.e., neither long term costs nor further acceleration, Table 2).

Gender and the benefits of working group participation

Given that WGs indeed catalyze researchers’ publication impacts, the next important question is whether this benefit is similar for men and women. Here we ask: Does WG participation magnify or ameliorate women’s H-index disadvantage? To assess this, we interact gender with the time intervals before and after the first WG experience (Table 1, Model 2).

Consistent with our earlier analysis (Fig. 1b), before the WG experience, women have significantly lower H-index progression than men, with women’s H-index predicted to in-

crease by 1.08 per year and men’s H-index by 1.52 per year (Fig. 3b). After their first WG, the H-index progression of both women and men accelerates. However, there is no significant interaction between gender and WG participation, meaning that this acceleration is similar for men and women (Table 2, Fig. 3b).

Furthermore, our survey suggests perceptions of indirect benefits from WGs are also similar for both women and men, with 78% and 80%, respectively, reporting benefits in addition to publications from their most recent WG (Fig. 2c). The majority of respondents reported that they developed new collaborations in their WG which carried forward into new projects. Roughly a quarter of respondents reported that their participation in a WG enabled them to reuse a database developed in a WG for a new purpose, and a quarter reported that their WG participation led to funding opportunities. Importantly, there was no gender difference in any of these proportions, suggesting that men and women perceive similar future benefits, at least for their most recent WGs (all tests: $\chi^2_1 < 0.18$, $p > 0.67$).

Citation rates of working group publications

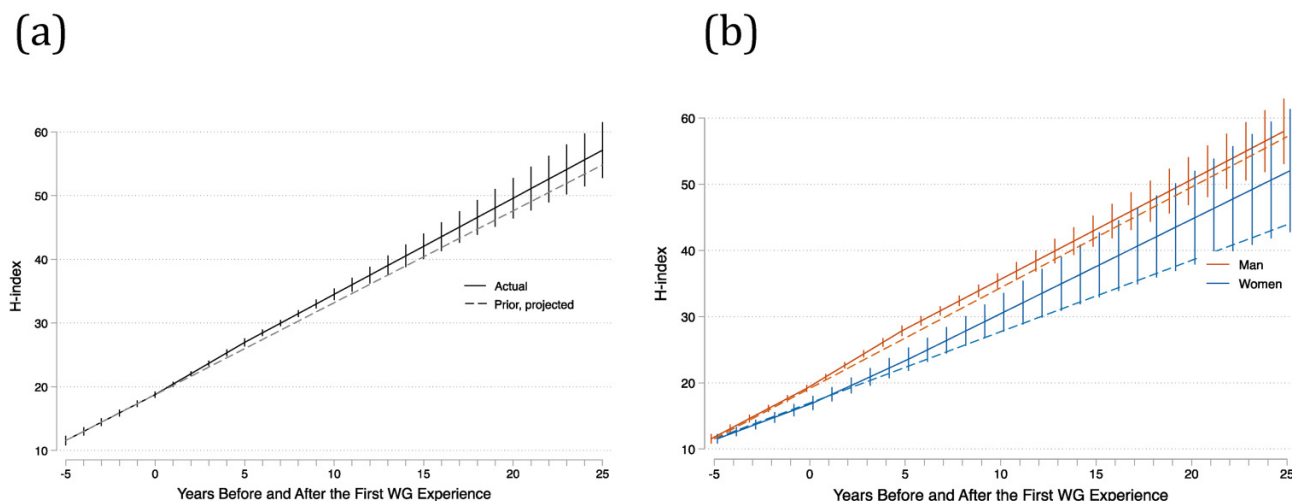
The immediate effects of WGs on H-indices may be related to the high citation rates of WG publications. Our analysis found that WG publications have on average $4.4\times$ more citations than those produced outside of WGs (Fig. 4). We found that this high citation rate appears to originate both from the type of research (synthesis vs. primary) commonly produced by WGs (>97% of WG publications are synthetic) as well as the actual WG method. Specifically, synthesis science—whether mathematical, conceptual, or statistical—is more highly cited than primary research (Figure 4; $\chi^2 = 35.3$, $p = 10^{-9}$). Independent of the type of research, publications from WGs are cited more than publications based on other, more traditional, methods (Fig. 4; WGs: $\chi^2 = 4.9$, $p = 0.03$). Although, we had only a few examples of primary research produced by WGs in our database (largely from globally distributed experiments or original data collection on the scientific process), these also tended to be more cited than primary research from more traditional methods (WG $\times$ research type interaction, $\chi^2 = 0.62$, $p = 0.43$).

Discussion

This study explores the impact of WGs—an effective approach to doing synthesis science—and gender on the H-index advancement of Canada’s ecology and evolution faculty. By combining analyses of researcher citations with a survey of the target population, several interesting findings emerged. First, participation in WGs leads to a 14% increase in the rate of H-index growth within 5 years of the first WG publication, and this higher rate is maintained subsequently. Second, despite marked gender disparities in representation and publication impact among Canada’s ecology and evolution faculty, we found no evidence of gender disparities in either participation or benefits from WGs.

There are several possible reasons for why WGs accelerate the H-indices of researchers. The most direct reason is that publications from WGs are typically more cited than other

Fig. 3. (a) On average, the H-index accrual rate increases 14% following the first working group publication of a researcher and remains at this elevated rate for at least the next 10 years. (b) This effect of working group experience on H-index accrual rates occurs for both men and women, with no significant gender difference in this effect. In both (a) and (b), the H-index accrual rate prior to the first working group publication is projected forward in time (“prior, projected”) to provide a visual counterfactual.



publications in ecology and evolution. In our analysis, synthesis research publications from WGs were cited $4.4\times$ more than primary research publications produced through other means. This estimate is slightly higher than a previous analysis of synthesis science publications from WGs which showed that, depending on the subject, such publications were cited $1\text{--}3\times$ more than the average publication (Halpern et al. 2020). This slight discrepancy may be because synthesis science publications were not removed from the pool of non-WG publications in the analysis by Halpern et al. (2020). By contrast, our analysis separates research type (synthesis vs. primary) from method (WG collaboration vs. traditional) and so is able to show that the high impact of WG publications is due to both aspects of WGs.

Previous analyses of ecology publications have also shown that longer papers with more authors tend to be cited more, with potential mechanisms including a greater diversity of ideas, greater findability, and higher self-citations (Fox et al. 2016). As WG publications are often substantial and multi-authored, similar mechanisms may underlie our finding that they are cited more than traditionally produced publications. Although we did not attempt to attribute mechanisms, we note that Halpern et al. (2020) demonstrated that the greater citation rate of WG publications was independent of author number. If the high impact of WG publications drives the acceleration of researchers' H-indices, then the benefit in co-authoring such publications must exceed the cost in terms of foregoing other research opportunities.

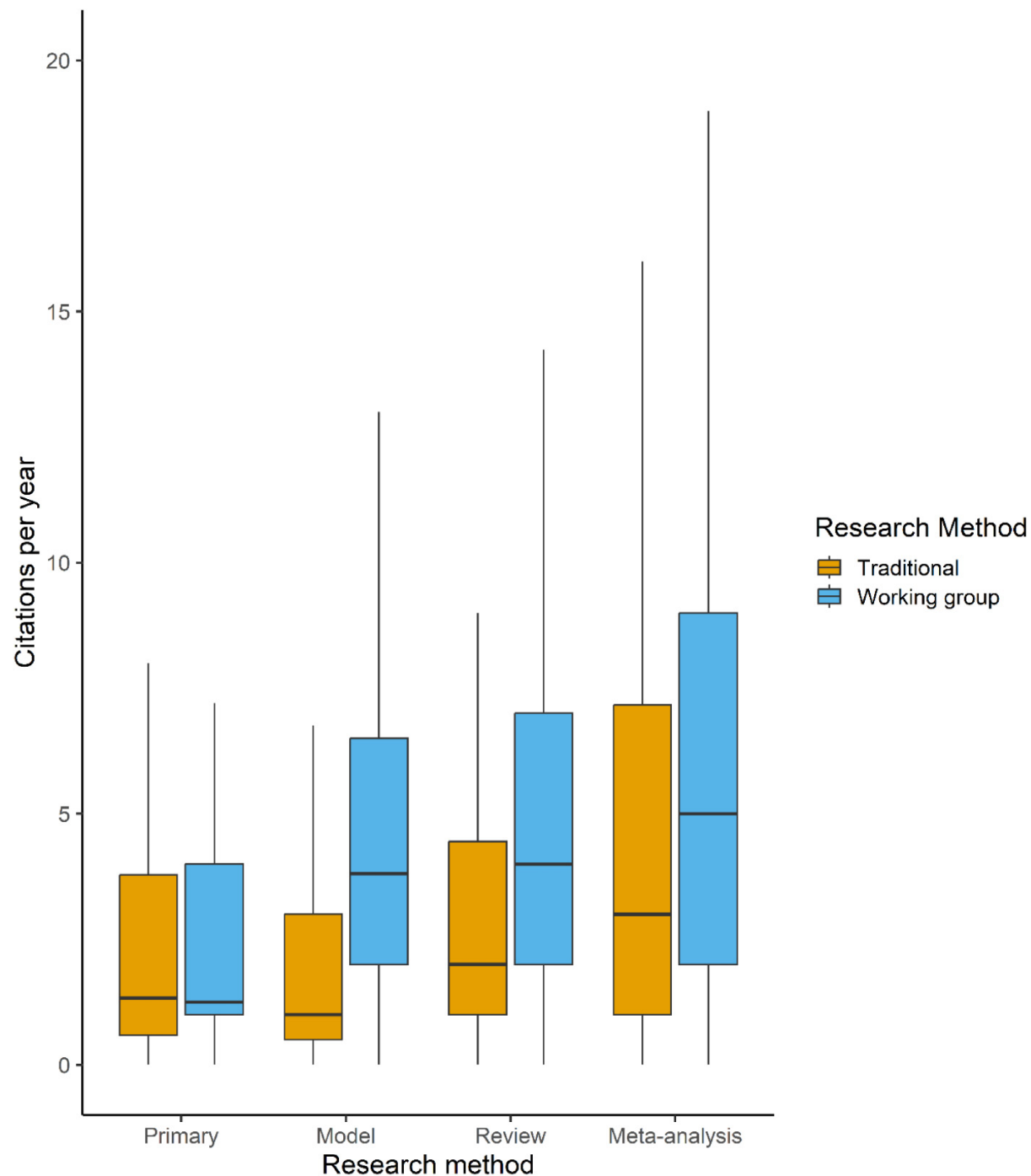
It is possible that this benefit may not only be the result of higher impact of WG publications, but also reflects greater efficiency of WGs in creating publications. For example, WGs often result in multiple publications that differ in the lead author, creating the potential for efficient sharing of the scientific workload. The composition of WGs may also lead to greater productivity. Many WGs are funded through synthe-

sis center competitions which favour groups assembled from a diversity of different institutions, disciplines, and genders. As both institutional and gender diversity are positively associated with the number or impact of publications produced by WGs, there is a positive selection for productive teams (Hampton and Parker 2011; Campbell et al. 2013).

We also evaluated the potential for WGs to indirectly accelerate H-index progression by enabling research networks, databases, or funding opportunities that facilitated new projects. In our survey, most WG participants reported such indirect benefits, particularly in terms of research networks. This corroborates an earlier analysis of WG participants which reported an increase in the number of co-authors following their first WG experience (Hampton and Parker 2011). It is possible that such indirect benefits, depending on their timing, contributed to the acceleration of H-indices subsequent to WG participation.

These effects of WGs occur in the context of profound gender disparities in the representation and H-index growth rates of Canadian faculty in ecology and evolution. Such gendered disparities are not unique to Canada, or to ecology and evolution. Many disciplines report that women are underrepresented among university faculty and have lower research productivity and impact (Astegiano et al. 2019; Huang et al. 2020). In Canada, women comprised only 42.1% of all university faculty in 2022 compared to 47.4% of the Canadian labour force (Statistics Canada 2023; World Bank 2023). Such gender inequities have many contributing factors, including systemic disadvantages faced by women in funding, institutional expectations, opportunities for collaboration, experiences of harassment and mistreatment, and domestic workloads (Bornmann et al. 2007; McLaughlin et al. 2012; Weisshaar 2017; Astegiano et al. 2019). Our study was not designed to parse such mechanisms. Instead, we ask if the disadvantages faced by Canadian women faculty in their research

Fig. 4. Publications vary in the average number of citations per year according to research type (primary research or synthesis research, with the latter comprising mathematical synthesis, conceptual synthesis, and statistical synthesis) and research method (working group or traditional). We consider any research method that is not a working group as traditional.



careers as a whole are also reflected in WGs. Here, the answer is no: the gender of researchers does not predict whether they will be selected for a WG (after taking into account their prior H-index and academic age), nor how much their H-index accrual increases after their first WG publication. Even when we examine WG participation and benefits in more detail, in our survey, we found no gender differences in rates of participation, of declining invitations or of receiving indirect benefits.

The lack of gender-bias in WG participation may reflect the policies of many, although not all, synthesis centers in either mandating or encouraging a minimum participation rate of women in funded WGs. We do caution that our analysis can only show that WG participation rates are equal (similar between men and women conditional on their H-index), not that they are equitable (similar between men and women conditional on their abilities as researchers), as the H-index may be a gender-biased metric of researcher quality.

Even if participation rates in WGs are gender-neutral, this does not guarantee that WG career benefits will also be gender-neutral; for example, not all participants may be included as authors in all publications. However, our longitudinal analysis and survey both suggest that such benefits are similar for women and men. It is remarkable that women have been able to achieve such parity in WGs while facing disadvantages in other research activities, such as promotion and funding (Astegiano et al. 2019). We do note that our analysis, by its narrow focus on H-indices, does not consider whether gender matters for WG's relationship with other metrics of academic achievement, such as the prestigious first, last and corresponding authorship positions on publications. Previous analyses have shown that women are systematically disadvantaged in terms of authorship position (Broderick and Casadevall 2019), including in ecology and evolution (Fox et al. 2018).

One of the strongest predictors of WG participation is the prior H-index of researchers. The funding of WGs is competitive and includes the curriculum vitae of participants (especially the WG leads). Such funding is needed to cover the substantial travel, accommodation and meal expenses of WGs. There is therefore a strong selective pressure on the H-indices of WG participants. Given that participation in WGs results in an acceleration of researcher H-index, there is a potential for positive feedback such that WG participation results in higher H-indices which enables greater WG participation. Researchers that participate in WGs also learn modes of collaborating particular to WGs and this, combined with their personal benefits realized from WG participation, may also result in continued use of this research mode. In our survey, 85% of respondents with WG experience had participated in multiple WGs. Informally, many synthesis centers report that the same researchers are often in multiple WGs, supporting such a mechanism (pers. comm. International Synthesis Consortium). One limitation of our study is that we considered only the effects of researchers' first WG experience, not the cumulative effect of subsequent WGs, but this would be a rich area for further study.

Our findings have several implications for science policy. First, we provide evidence that the synthesis science carried out by WGs not only benefits science (Halpern et al. 2020) but also scientists. This alignment of interests is a prerequisite for the long-term sustainability of the method. However, sustainability is also contingent on funding mechanisms for WGs, such as financial support of synthesis centers (Baron et al. 2017), and the benefits outlined in this study provide a rationale for such continued funding. Second, the parity demonstrated between men and women in WG participation and benefits stands out against a backdrop of relentless gender disparity in science. One policy mechanism to offset this disparity is therefore to fund structured collaborations like WGs that require diverse teams. Just as gender-diverse corporate boards are more profitable (Dăng et al. 2020), there is evidence that gender diversity in WGs leads to higher impact publications (Campbell et al. 2013). Third, given the benefits of WG participation to academic careers, it is essential that the skills and opportunities for WG participation be equitably and widely available. Specifically, we need widespread train-

ing of graduate students and postdoctoral researchers in the collaboration and data science skills required for WGs, such as by the CIEE's Living Data Project (Bledsoe et al. 2022). Finally, although we were restricted by data availability to compare WG benefits between men and women, there is an urgent need to examine equity in terms of other demographics such as race, Indigenous status, (dis)ability, sexual orientation, and other gender identities. Collection and equity analysis of such data should be a priority for synthesis centers.

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Data availability

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Competing interests

The authors declare there are no competing interests.

Supplementary material

Supplementary data are available with the article at <https://doi.org/10.1139/facets-2024-0143>.

References

- Abramo, G., D'Angelo, C.A., and Di Costa, F. 2009. Research collaboration and productivity: is there correlation? [journal article]. *Higher Education*, **57**(2): 155–171. doi:10.1007/s10734-008-9139-z.
- Allison, P.D. 2009. Fixed effects regression models. SAGE publications.
- Astegiano, J., Sebastián-González, E., and Castanho, C.d.T. 2019. Unraveling the gender productivity gap in science: a meta-analytical review. *Royal Society Open Science*, **6**(6): 1–12.
- Baron, J.S., Specht, A., Garnier, E., Bishop, P., Campbell, C.A., Davis, F.W., et al. 2017. Synthesis Centers as critical research infrastructure. *Bioscience*, **67**(8): 750–759. doi:10.1093/biosci/bix053.
- Bledsoe, E.K., Burant, J.B., Higinio, G.T., Roche, D.G., Binning, S.A., Finlay, K., et al. 2022. Data rescue: saving environmental data from extinction. *Proceedings of the Royal Society B Biological Sciences*, **289**(1979): 20200938. doi:10.1098/rspb.2022.0938. PMID: 35855607.
- Bordons, M., Gómez, I., Fernández, M.T., Zulueta, M.A., and Méndez, A. 1996. Local, domestic and international scientific collaboration in Biomedical research [journal article]. *Scientometrics*, **37**(2): 279–295. doi:10.1007/bf02093625.
- Bornmann, L., Mutz, R., and Daniel, H.-D. 2007. Gender differences in grant peer review: a meta-analysis. *Journal of Informetrics*, **1**(3): 226–238. doi:10.1016/j.joi.2007.03.001.
- Bozeman, B., and Corley, E. 2004. Scientists' collaboration strategies: implications for scientific and technical human capital. *Research Policy*, **33**(4): 599–616. doi:10.1016/j.respol.2004.01.008.
- Broderick, N.A., and Casadevall, A. 2019. Gender inequalities among authors who contributed equally. *Elife*, **8**, e36399. doi:10.7554/eLife.36399. PMID: 30698140.
- Campbell, L.G., Mehtani, S., Dozier, M.E., and Rinehart, J. 2013. Gender-heterogeneous working groups produce higher quality science. *PLoS ONE*, **8**(10): e79147. doi:10.1371/journal.pone.0079147. PMID: 24205372.
- Castilla, E.J., and Benard, S. 2010. The paradox of meritocracy in organizations. *Administrative Science Quarterly*, **55**(4): 543–676. doi:10.2189/asqu.2010.55.4.543.
- Đặng, R., Houanti, L.H., Reddy, K., and Simioni, M. 2020. Does board gender diversity influence firm profitability? A control function approach. *Economic Modelling*, **90**, 168–181. doi:10.1016/j.econmod.2020.05.009.
- Dobbin, F., and Kalev, A. 2016. Why diversity programs fail [Article]. *Harvard Business Review*, **94**(7/8): 52–60. <https://stanford.idm.oclc.org/login?url=https://search.ebscohost.com/login.aspx?direct=true&db=bth&AN=116330233&site=ehost-live&scope=site>.
- Fox, C.W., Paine, C.E.T., and Sauterey, B. 2016. Citations increase with manuscript length, author number, and references cited in ecology journals. *Ecology and Evolution*, **6**(21): 7717–7726. doi:10.1002/ece3.2505.
- Fox, C.W., Ritchey, J.P., and Paine, C.E.T. 2018. Patterns of authorship in ecology and evolution: first, last, and corresponding authorship vary with gender and geography. *Ecology and Evolution*, **8**(23): 11492–11507. doi:10.1002/ece3.4584.
- Gazni, A., and Didegah, F. 2011. Investigating different types of research collaboration and citation impact: a case study of Harvard University's publications [journal article]. *Scientometrics*, **87**(2): 251–265. doi:10.1007/s11192-011-0343-8.
- Halpern, B.S., Berlow, E., Williams, R., Borer, E.T., Davis, F.W., Dobson, A., et al. 2020. Ecological synthesis and its role in advancing knowledge. *Bioscience*, **70**(11): 1005–1014. doi:10.1093/biosci/biaa105.
- Hampton, S.E., and Parker, J.N. 2011. Collaboration and productivity in scientific synthesis. *Bioscience*, **61**(11): 900–910. doi:10.1525/bio.2011.61.11.9.
- Hampton, S.E., Strasser, C.A., Tewksbury, J.J., Gram, W.K., Budden, A.E., Batcheller, A.L., et al. 2013. Big data and the future of ecology. *Frontiers in Ecology and the Environment*, **11**(3): 156–162. doi:10.1890/120103.
- Huang, D.-w. 2015. Temporal evolution of multi-author papers in basic sciences from 1960 to 2010. *Scientometrics*, **105**(3): 2137–2147. doi:10.1007/s11192-015-1760-x.
- Huang, J., Gates, A.J., Sinatra, R., and Barabási, A.-L. 2020. Historical comparison of gender inequality in scientific careers across countries and disciplines. *Proceedings of the National Academy of Sciences*, **117**(9): 4609–4616. doi:10.1073/pnas.1914221117.
- Kang, S.K., DeCelles, K.A., Tilcsik, A., and Jun, S. 2016. Whitedened résumés: race and self-presentation in the labor market. *Administrative Science Quarterly*, **61**(3): 469–502. doi:10.1177/0001839216639577.
- Koltun, V., and Hafner, D. 2021. The h-index is no longer an effective correlate of scientific reputation. *PLoS ONE*, **16**(6): e0253397. doi:10.1371/journal.pone.0253397.
- Larivière, V., Gingras, Y., Sugimoto, C.R., and Tsou, A. 2015. Team size matters: Collaboration and scientific impact since 1900. *Journal of the Association for Information Science and Technology*, **66**: 1323–1332.
- Lee, S., and Bozeman, B. 2005. The impact of research collaboration on scientific productivity. *Social Studies of Science*, **35**(5): 673–702. doi:10.1177/0306312705052359.
- Leimu, R., and Koricheva, J. 2005. Does scientific collaboration increase the impact of ecological articles? *BioScience*, **55**: 438.
- McLaughlin, H., Uggen, C., and Blackstone, A. 2012. Sexual harassment, workplace authority, and the paradox of power. *American Sociological Review*, **77**(4): 625–647. doi:10.1177/0003122412451728.
- Miranda, R., and Garcia-Carpintero, E. 2018. Overcitation and overrepresentation of review papers in the most cited papers. *Journal of Informetrics*, **12**(4): 1015–1030. doi:10.1016/j.joi.2018.08.006.
- Misra, J., Smith-Doerr, L., Dasgupta, N., Weaver, G., and Normanly, J. 2017. Collaboration and Gender Equity among Academic Scientists, **6**(1): 25. <https://www.mdpi.com/2076-0760/6/1/25>.
- Rajan, R.G., and Subramanian, A. 2008. Aid and growth: what does the cross-country evidence really show? *Review of Economics and Statistics*, **90**(4): 643–665. doi:10.1162/rest.90.4.643.
- Rossiter, M.W. 1993. The Matthew Matilda effect in science. *Social Studies of Science*, **23**(2): 325–341. doi:10.1177/030631293023002004.
- Sarsons, H. 2017. Recognition for group work: gender differences in academia. *American Economic Review*, **107**(5): 141–145. doi:10.1257/aer.p20171126.
- Stainback, K., and Tomaskovic-Devey, D. 2009. Intersections of power and privilege: long-term trends in managerial representation. *American Sociological Review*, **74**(5): 800–820. doi:10.1177/000312240907400506.
- Statistics Canada. 2023. Statistics on full-time academic teaching staff at Canadian universities. Available from <https://www150.statcan.gc.ca/n1/pub/71-607-x/71-607-x2019027-eng.htm>.
- Tomaskovic-Devey, D., and Avent-Holt, D. 2019. Relational inequalities: an organizational approach. Oxford University Press, USA.
- Uhly, K.M., Visser, L.M., and Zippel, K.S. 2017. Gendered patterns in international research collaborations in academia. *Studies in Higher Education*, **42**(4): 1–23. doi:10.1080/03075079.2015.1072151.
- Weisshaar, K. 2017. Publish and perish? An assessment of gender gaps in promotion to tenure in academia. *Social Forces*, **96**(2): 529–560. doi:10.1093/sf/sox052.
- Wooldridge, J.M. 2016. Introductory econometrics: a modern approach. 6th ed. Cengage Learning.
- World Bank. 2023. World development indicators. The World Bank.
- Wuchty, S., Jones, B.F., and Uzzi, B. 2007. The increasing dominance of teams in production of knowledge. *Science*, **316**(5827): 1036–1039. doi:10.1126/science.1136099.

Replacing the Canadianization Generation: An Examination of Faculty Composition from 1977 through 2017

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Abstract

Amid growing numbers of doctoral graduates entering an increasingly competitive global academic job market, concerns about equity in the hiring process and the value of the Canadian Ph.D. are mounting. Grounded within the historical context of the Canadianization Movement, we examine the doctoral credentials of 4,934 U15 social science faculty between 1977 and 2017 to understand the ebb and flow of incoming and outgoing faculty across the country's academic field. Our trend analyses reveal an overall increase in the proportion of Canadian-trained faculty hires with the noted exceptions of Canada's top three universities who display a strong presence of high-status American-trained faculty throughout. Results from the contemporary period, between 1997 and 2017, reveal a time of retirement during which outgoing Canadian-trained faculty are replaced with increasing proportions of American-trained academics.

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Résumé

Pendant qu'un nombre croissant de doctorants s'insèrent dans un marché du travail académique global de plus en plus compétitif, des interrogations concernant l'équité dans les processus d'embauche et la valeur des doctorats canadiens émergent. En tenant compte d'un contexte historique lié au Mouvement de Canadianisation, nous examinons les diplômes de doctorat de 4,934 'U15' professeurs en sciences sociales entre 1977 et 2017 afin de comprendre le flot fait d'arrivés et de départs de professeurs à travers le champ académique du pays. Nos analyses révèlent une augmentation générale de la proportion des professeurs formés au Canada, à l'exception notable des trois universités de plus haut niveau, qui ont une grande présence de professeurs de statuts élevés formés aux États-Unis. Les résultats de la période contemporaine, soit entre 1997 et 2017, révèlent une ère de retraite pendant laquelle des professeurs canadiens qui partent sont remplacés par une proportion croissante de collègues formés aux États-Unis.

IN A RECENT UNIVERSITY Affairs piece, Maren Wood asked, “What is the value of a Canadian PhD to universities in Canada?” (Wood 2017). Amid increasing numbers of doctoral enrollments (from 29,874 in 2003 to 46,782 in 2011; Looker 2015), a rising “underclass” (Rajagopal 2002) of sessional instructors (Field et al. 2014), and Ph.D. aspirations fixed on the tenure track (Desjardins 2012), it is a pressing question in need of answers. Responses to the job market “crisis” have often singled out the current (or coming) mass retirement of Canadian faculty as the solution that will open countless new positions for our best and brightest Canadian candidates. Many others, however, are skeptical. As research-intensive schools in Canada engage further in the global quest for scientific “excellence” (Gingras 2010; Stack 2016) and interest from international “post-docs and junior faculty right through to mid-career, to truly established stars who want to move here” (Blackwell 2017) continues to increase, the degree to which professorial positions will be filled by domestic appointments remains to be seen. While this wave of interest from foreign talent, aligned with the Federal government science policy, might be due to political instability in the United Kingdom and the United States, the current situation recalls many of the anxieties shared by Canadian academics during a pivotal moment in the postwar history of the country’s academic field: the Canadianization Movement.

Employing original longitudinal data spanning from 1977 through 2017, we aim to document the movements of external domination vis-à-vis nationalizing forces within the Canadian social sciences. Specifically, we look at the national Ph.D. origin of 4,934 social scientists from five disciplines—anthropology, economics, political science, psychology, and sociology—working at 15 research-intensive Canadian universities (U15) during this period. We pursue two research objectives. The first is

historical in scope and focuses on the period from 1977 to 1997. We investigate the composition patterns of incoming and outgoing faculty at four substantively distinct classes of university. We focus on the doctoral Canadianization and de-Americanization over time, considering why domestic credentials did not take root as valued symbolic capital at McGill, University of Toronto (U of T), and the University of British Columbia (UBC). We propose two influential field-wide forces that can help us understand the faculty reproduction practices of these three schools: the influence of networks and faculty majority, and principles of scholastic merit and prestige. The second objective focuses on the contemporary period from 1997 through 2017. Here, we examine the differences between incoming and outgoing faculty for each class with marked attention to two field-wide forces that resonate during this period: the cycle of faculty renewal (retirement and hiring), and the departure of the Canadianization generation amid an emerging global academic market. This exploratory examination of overall, incoming, and outgoing faculty composition trends over the past 40 years can facilitate a richer understanding of the value of the Canadian Ph.D. and allow us to consider larger unchecked questions about the academy's de-Canadianization over time.

POSTWAR ACADEMIC AMERICANIZATION AND THE CANADIANIZATION MOVEMENT

Marked by the crossing of more than 50,000 Vietnam war draft dodgers (Hagan 2001) and a burgeoning Canadian postsecondary education system forced to hire from outside the country (Axelrod 1982; Cormier 2004), the 1960s and 1970s ushered in the single largest movement of U.S. academics in Canadian history (Brown 1967). In response to the perceived overtaking of Canadian academe by foreign academics, the Canadianization Movement emerged as a call for the creation and integration of Canadian content in universities' and colleges' teaching materials (Symons 1975), and action against the alleged exclusion of Canadian graduates from tenure track jobs at Canadian universities (Cormier 2004). Sparked in part by the empirical conclusions presented in their book *The Struggle for Canadian Universities*, Mathews and Steele (1969) note that between 1961 and 1968 the number of Canadians working at Canadian universities had decreased by 25 percent and two-thirds of faculty had obtained their first degree outside Canada. These two nationalist Canadians saw a clear connection between the domination of the country's higher institutions of knowledge and culture and "a much larger colonial problem" Canada would not admit (Mathews 2014:1). Using country of first degree as indicator of citizenship, they concluded that more than 66 percent of Canadian professoriate were non-Canadian. Although this finding was based on inferred citizenship—a measure prone to conflate Canadian-born citizens who study outside of Canada with non-Canadians—and likely exaggerated the proportion of

national and foreign faculty at Canadian universities, it acted as a powerful political agent for social mobilization and Canadian nationalism. In the decade that followed, Canadian scholars, politicians, and educational administrators, as well as other decision makers, became increasingly wary of the number of foreign faculty teaching at Canadian universities and of the lack of Canadian-specific curriculum content (Axelrod 1982:168–70).

The presence of American academics in the 1960s and 1970s emerged in part from the explosive growth of the Canadian postsecondary education system and the dire need for qualified instructors (Brym 2002; Williams 2005). Between 1960 and 1975, the number of full-time university instructors increased fourfold, from 7,760 to more than 30,000 (Dandurand 1989:68); many of the disciplines simply did not have the programs and resources to train and produce graduates and Ph.D. students. This was especially the case in relatively young social science disciplines such as anthropology and sociology; however, the shortage crisis was not limited to these disciplines. For instance, in the 1960s, Canadian universities conferred a total of only 7,059 doctoral degrees (Statistics Canada 1973), while during that decade American schools awarded 16,284 (Chiswick, Larsen, and Pieper 2010:3).

Following the publication of Mathews and Steele's (1969) book, Statistics Canada collected citizenship data of professors working in the country for the first time. Measuring professoriate's "synchronic" citizenship (i.e., citizenship at the time of the survey), figures revealed that the proportion of professors in Canada who were citizens totaled 62.6 percent in 1970 to 1971 and 67.5 percent in 1974 to 1975. Although limited, these national-level statistics alluded to a dominant presence of Canadian *citizens* in the professorial rank since at least the late 1960s, findings that called the perceived American dominance into question. As noted by Fischer (1991), however, "the social science vacancies were filled by relatively young new PhD graduates from the US" (p. 80). Similarly, Cormier (2004:193) found that in 1970, Canadian sociology and anthropology departments consisted of only 39 and 29 percent Canadian faculty, respectively.

SYMBOLIC STRUGGLES AND THE CANADIAN FIRST POLICY (1970s AND 1980s)

Given social sciences' state of disproportional external domination, it is not surprising that these disciplines played a role in pushing the domestication agenda forward during that period. The path toward academic sovereignty was marked by a controversial Canadian Sociology and Anthropology Association policy recommendation that encouraged prioritized hiring of Canadian citizens and urged that at least 75 percent of graduate students be Canadian (Cormier 2004:127). Reception to the recommendation was contentious, spurring debates over identity politics, hiring procedures, and pedagogical issues (see Cormier 2004:125–33; Mathews 2014) that set in

motion the process of establishing Canadian citizenship and Canadian credentials as the dominant form of symbolic capital in the domestic academic field. This is not to say that more classic forms of scientific capital (e.g., publications, prizes, or awards) were insignificant, in fact, divisive lines were drawn between those who defended a merit-based hiring structure regardless of citizenship or place of study, and those who questioned the balance between merit and equity inherent to academic hiring practices.

These symbolic struggles for the recognition of “Canadian-ness” as a legitimate form of scientific capital culminated in 1981 when the Federal government adopted the *Canadian First Policy*, a two-tier advertising, recruitment, and interviewing process that gave preference to Canadian citizens and landed immigrants during the first round of nationwide academic job postings. Scholars often mention these points only in passing, such as Nakhaie (2008), who claims that “the Canadianization Movement of the 1970s, which intended to limit access of non-Canadians to jobs in the academy, may have reversed this pattern” (p. 363) of Americanization. The general assumption is that, overtime, the effects of the Canadian First Policy and the growth of Canadian universities’ doctoral production translated to the substantial increase of both Canadian citizens and/or Canadian-trained Ph.D.s holding faculty positions. More recent Statistics Canada numbers, which show that in 2006 to 2007 86.8 percent of professors in the country were Canadian citizens, seem to corroborate this narrative (Teichler, Arimoto, and Cummings 2013:84). But what about faculty’s national Ph.D. origin? Although the Canadian First Policy did not officially include a provision for the preferential hiring of Canadian-trained candidates, it is well recognized—or at the very least assumed in the literature—that the policy facilitated the nationalization of Canadian academics’ doctoral origin (Gingras 2010; Warren 2014; Wilkinson et al. 2013).

CURRENT STRUGGLES: CANADIAN ACADEME AND INTERNATIONAL FIELDS (2003–2017)

The contemporary period from the late 1990s onward is marked by mass faculty retirement and replacement, notable changes to policies regarding faculty mandatory retirement (Worswick 2005), revisions to the Canadian First Policy guidelines (Tremblay, Hardwick, and O’Neill 2014), further entrenchment of internationalization strategies (Gingras 2010), and the emergence of global university rankings as powerful verticalizing technologies (Stack 2016). This move toward internationalization is described by Gingras (2010) as the new dominant rhetoric that has come to equate “excellence” with “international,” a trend tending to favor the hiring of foreign scholars. Accordingly, as this argument suggests, international forms of scientific capital such as international experience, publication and

research record, and doctoral credentials established themselves further as dominant forms of scholastic currency at the expense of domestic ones.

Where empirical efforts examining academic citizenship were triggered by a perceived postwar Americanization—or “de-Canadianization” in a context of Ph.D. shortage—investigations into the national origin of professoriate credentials were prompted by more recent claims of re-Americanization and internationalization of the country’s academic and higher educational fields in a context of Ph.D. “overproduction.” This body of research, though fragmented and rarely longitudinal in scope, suggests somewhat contradictory trends regarding the national origins of faculty’s Ph.D. training. Working with data on the social sciences from 2005, Gingras (2010) found that 55 percent of Canadian-trained faculty employed at 10 of the country’s largest universities held doctorates from Canadian universities. Noting a proportional decrease of Canadian-trained professoriate since its peak in the mid-1990s, he states that we are seeing the “end of the Canadianization Movement” as a result of globalization forces shaping the domestic field (Gingras 2010; for similar trends in philosophy, see also Groarke and Fenske 2009). Running contrary to this observation, Wilkinson et al. (2013) surveyed *assistant professors* employed in all 56 of Canada’s sociology departments in 2012 and found that only five departments had fewer than 50 percent Canadian-trained faculty (for similar trends, also see McLaughlin 2005, 2006; Tremblay et al. 2014). Although the source of these discrepancies could be methodological in nature, these reports all seem to agree that larger research-intensive universities tend to hire more foreign-trained Ph.D.s.

RESEARCH OBJECTIVES

In recent years, perception of the Canadian academic field as a “flat system” that encourages “egalitarianism over hierarchy” (Davies and Hammack 2005) is challenged by mounting evidence of field-wide scholastic stratification cut along lines of gender (Bell and Gordon 1999), race (Nakhaie 2008), language (Warren 2014), research funding (Guppy, Grabb, and Mollica 2013; Siler and McLaughlin 2008), research “quality” and publication record (Gingras 2010), and institutional prestige networks (Goyder 2009). What is lacking is a clear understanding of how doctoral origin contributes to inequities on the Canadian academic job market over time. Our time-series analyses examine the differential rise and more recent fluctuations of Canadianization efforts across differently positioned universities in the national academic field since the late postwar era. Our lines of inquiry examine the following: (1) the overall proportional variation of Canadian-trained, American-trained, U.K.-trained, and other internationally trained faculty over key moments in the history of the Canadianization Movement; (2) the proportional balance between incoming and outgoing faculty; and

(3) the representation of high-status American universities in faculty composition.

DATA

The time-series analyses employ data on academic rank and the names of institutions that issued bachelor's, master's, and doctoral degrees to faculty working at Canadian social science departments from the late 1970s (prior to the implementation of the Canadian First Policy) through 2017. The first stage of data collection utilized hard copy, microfiche, and digital forms of academic calendars to identify the complete name, the academic rank (assistant, associate, or professor), and the name of the university where the faculty member had received his or her degree. For instances of incomplete or missing information, the Internet archive, ProQuest dissertation database, university promotional archives, and faculty publications were used to recover missing data.

Criteria for sampling Canadian universities and social science departments considered their status as Ph.D.-granting and research-intensive universities with active and distinct departments between 1977 and 2017. These criteria informed the selection of universities that are part of the U15 group, a pan-national network of Canada's top 15 research-intensive universities formed in 1985, that, as of 2016, accounted for 87 percent of contracted private-sector research, 80 percent of competitively allocated research funding, and 71 percent of Canadian full-time Ph.D. students (see Table 1 for list of U15 schools). Five social science departments that were active at each of the 15 universities between 1977 and 2017 were selected: anthropology, economics, political science, psychology, and sociology.

Finally, we collected data on academic training of faculty at 10-year intervals, starting in 1977 to 1978 and ending in 2016 to 2017. We identify and track 4,934 unique social scientists who were assistant, associate, and full professor faculty members in five departments at U15 schools between 1977 and 2017. Of the 4,934 faculty members, our online data sleuthing and cleaning processes resulted in the documentation of academic rank and name of academic institution of *highest conferred degree* for all the cases in our sample.

UNIVERSITY CLASSES

Informed by publicly available data on Canadian university metrics of Ph.D. production, sponsored research income, faculty size, international rankings, and theoretically informed criteria, we parse U15 schools into the following four substantively distinct classes (see Table 1).

Table 1
Faculty Counts per Collection Point

Classes	U15	1977 to 1978	1987 to 1988	1997 to 1998	2007 to 2008	2016 to 2017
Glocal class	University of Toronto	270	284	267	293	367
	University of British Columbia	134	130	124	142	180
	McGill University	95	105	113	147	143
	Total	499	519	504	581	690
National class	University of Alberta	133	156	128	121	122
	McMaster University	107	115	105	120	118
	University of Waterloo	108	116	109	119	120
	Western University	136	182	161	156	160
	Queen's University	108	126	116	110	87
	Total	592	694	619	626	607
Regional class	University of Calgary	105	109	97	116	148
	University of Manitoba	141	136	118	119	103
	University of Saskatchewan	77	89	84	84	80
	Dalhousie University	82	83	80	93	95
	Total	405	417	379	412	425
French class	University of Ottawa	94	98	95	142	170
	Laval University	118	126	159	96	123
	University of Montréal	140	155	173	153	162
	Total	352	379	426	391	456
Overall total		1848	2010	1929	2011	2178

*Glocal Class*¹

This class is formed of Anglo-Canada's top three research-intensive universities: McGill University, U of T, and UBC. These schools hold dominant positions in several annual global ranking scales (e.g., Times Higher Education, QS World Rankings) as well as national rankings (e.g., McLean's, Research Infosource). Further distinguishing factors include research-sponsored income, placing them at the top three in Canada for at least the last 20 years (Research Infosource 2017), and doctoral production (they awarded the highest number of Ph.D. degrees between 1999 and 2012; Looker 2015).

National Class

The second class consists of four Ontario-based universities—Western (formally Western Ontario), McMaster, Queen's, and Waterloo—and the University of Alberta. Behind the Glocal class, these five Ph.D.-granting schools rank highly on the national stage and hold higher ranking positions internationally than Regional class universities. They are also right behind the Glocal class in terms of research-sponsored income (Research Infosource 2017), doctoral production (Looker 2015), and university ranking position.

Regional Class

Located in less densely populated regions, Dalhousie, University of Calgary, University of Manitoba, and the University of Saskatchewan hold high status in their respective regional markets and tend to rank right behind National class universities in research-sponsored income (Research Infosource 2017), doctoral production (Looker 2015), and university rankings, though University of Calgary holds notably higher national position than the other three.

French Class

Finally, the French class consists of U15's only two French-speaking universities, Montréal and Laval, and Ontario's only bilingual university, Ottawa. Researchers often cast Québec's atypical characteristics as legitimate grounds to exclude French-speaking universities from their sampling strategies, focusing exclusively on the Anglophone portions of the field. The inclusion of French language universities is crucial for a better understanding of the whole national field. While French

¹ The term "Glocal" is loosely informed by sociologist Roland Robertson's (1995) concept of globalization, a portmanteau of "global" and "local."

class schools were not directly affected by, or as much engaged with, the Canadianization Movement, la Belle Province was undergoing its own historical political and cultural nationalist awakening during that time. That is to say, the Québec field was nevertheless affected by the regulations set out by the Canadian First Policy.

RESULTS

The first series of figures present the overall distribution (proportion) of countries where all tenure-track faculty (assistant, associate, and full professors) received their highest degrees within Glocal, National, Regional, and French university classes across all five collection points. National origin of highest degree is coded into four substantively relevant categories: Canada, the United States, the United Kingdom (England, Scotland, Wales, or Ireland²), and other nations (consisting of all remaining countries). The second series of figures (two through five) present the proportional distributions of national Ph.D. origins for *incoming* (new to a university between two time points) and *outgoing* (exited a university between two time points) faculty over time. Though presented together in the figures, incoming and outgoing proportions are independent from each other, each, respectively, amounting to 100 percent. Additionally, we present the raw counts and the difference between the number of incoming and outgoing faculty for each time period. Differing from Figure 1, these distributions capture more precisely the hiring trend patterns of university classes over time. The third and fourth analyses present lists of the 10 most successful schools in placing their Ph.D.s at each of the four university classes (Table 2) and the proportion of the top 15 high-status American universities³ that constitute the American-trained faculty for Glocal, National, and Regional classes in 1977 and 2017 (Table 3).

THE MOVEMENT OF CANADIANIZATION OVER TIME

The stratified nature of Canada's academic field of research-intensive universities is captured prominently in Figure 1, where each of the four U15 classes presents distinctive domestication trends. Perhaps most emphatically, the Glocal class displays a persistent ≈ 65 percent proportion of American-trained faculty composition *overall* (Figure 1), coupled with the proportional increase of American-trained faculty *hires* that rise from 50 percent in 1977 to 1987 to 68 percent in 2007 to 2017 (Figure 2).

² Though the United Kingdom includes only Northern Ireland, for simplicity's sake we include all Irish universities in our coding.

³ Top 15 American universities ranked by Times Higher Education in 2017. Also present in top 15 in 2011 and represented in Burris's (2004:247) top 25.

Figure 1

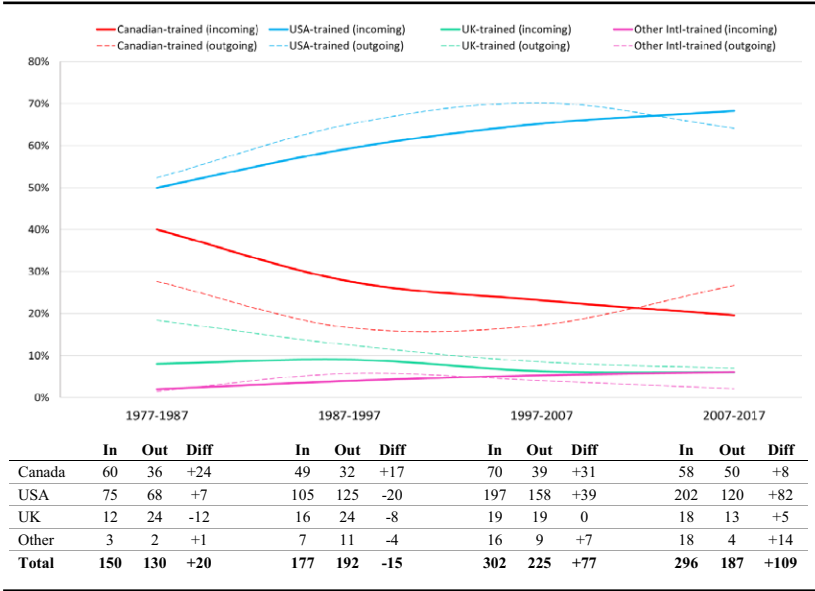
Faculty Credentials at U15 Universities Over Time [Color figure can be viewed at wileyonlinelibrary.com]



In terms of incoming faculty, Glocal schools' social science departments etched their most proportionately egalitarian Canadian-to-American hiring period between 1977 and 1987—around the time the Canadian First Policy was enacted—in which a narrow 10 percent difference separated the total proportion of Canadian-trained (40 percent; $n = 60$) and American-trained (50 percent; $n = 75$) hires (see Figure 2). This time of increased Canadian-trained scholars was short lived, as the hiring practices of Glocal schools shifted markedly after 1987. Around this time the proportion of American-trained faculty hires increased to 59.3 percent ($n = 105$) from 1987 to 1997, peaking at 68.2 percent ($n = 202$) during the most recent decade. These increases are mirrored by a decrease in the proportion of incoming Canadian-trained faculty, dropping to a low of 19 percent between 2007 and 2017.

Figure 2

Proportions and Raw Counts of Incoming and Outgoing Faculty at Glocal Universities [Color figure can be viewed at wileyonlinelibrary.com]

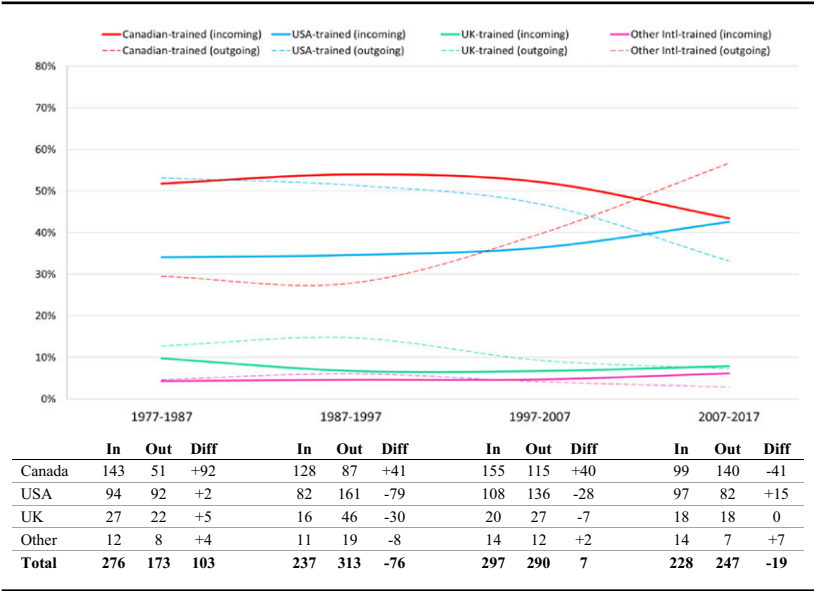


The National class, on the other hand, presents very different trends. Between 1977 and 1997, while these five schools recorded their highest increase to the *overall* proportion of Canadian-trained faculty (from 28 to 48 percent; Figure 1), the composition of *incoming* faculty converges over time, marked by decreasing proportions of Canadian-trained and increasing proportions of American-trained faculty hires, reaching parity in 2017 for the first time in at least three decades (see Figure 3). The near equal number of incoming Canadian-trained ($n = 99$) and American-trained ($n = 97$) faculty between 2007 to 2017 is underscored by a pivotal reversal of *outgoing* trends, where, for the first time since the 1977 to 1987 period, there are more Canadian-trained faculty leaving than entering this class. The proportion of Canadian-trained faculty who left National universities (57 percent; $n = 140$) is nearly double that of the American-trained faculty (33 percent; $n = 82$).

Regional schools' domestication trends show similarities with the National class up until 2007, registering the second largest increase in the

Figure 3

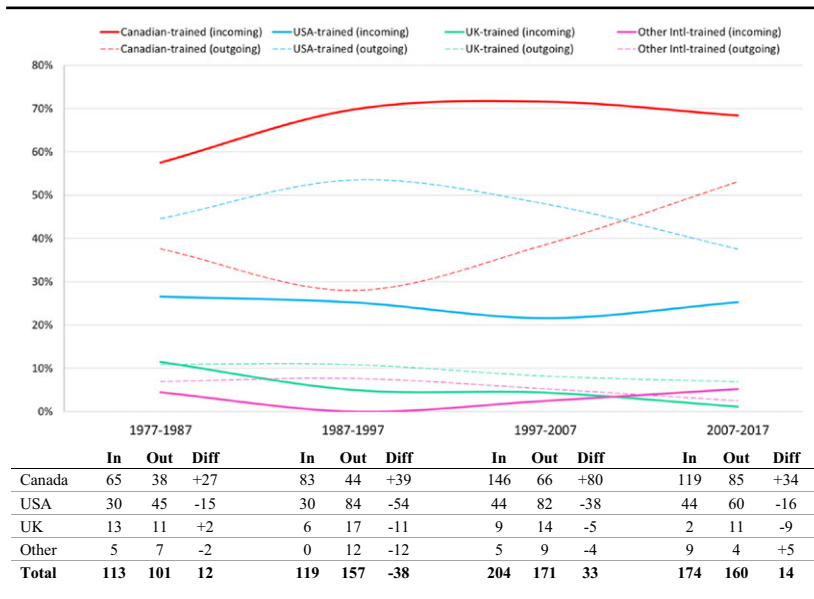
Proportions and Raw Counts of Incoming and Outgoing Faculty at National Universities [Color figure can be viewed at wileyonlinelibrary.com]



overall proportion of faculty holding domestic Ph.D.s (from 28 to 47 percent; Figure 1). For *incoming* faculty (Figure 4), the proportion of Canadian-trained hires increases from 58 percent to a high of 72 percent in 1997 to 2007, and drops slightly to 68 percent for the first time between 2007 and 2017. Proportions of incoming American-trained faculty, though decreasing slightly over time, are relatively stable near 25 percent over 40 years, reaching a low of 22 percent ($n = 44$) during the decade spanning 1997 to 2007. Underlying these trends are steep changes to the composition of *outgoing* faculty since the 1987 to 1997 period, where the proportion of Canadian-trained faculty leaving Regional schools increases from 28 to 53 percent in 2007 to 2017—surpassing the proportion of outgoing American-trained faculty for the first time in at least 40 years. Though smaller in numbers, it is worth noting the continued decrease in the proportion of U.K.-trained faculty hires over time, dropping incrementally from 12 percent ($n = 13$) in 1987 to 1 percent ($n = 2$) in 2017.

Figure 4

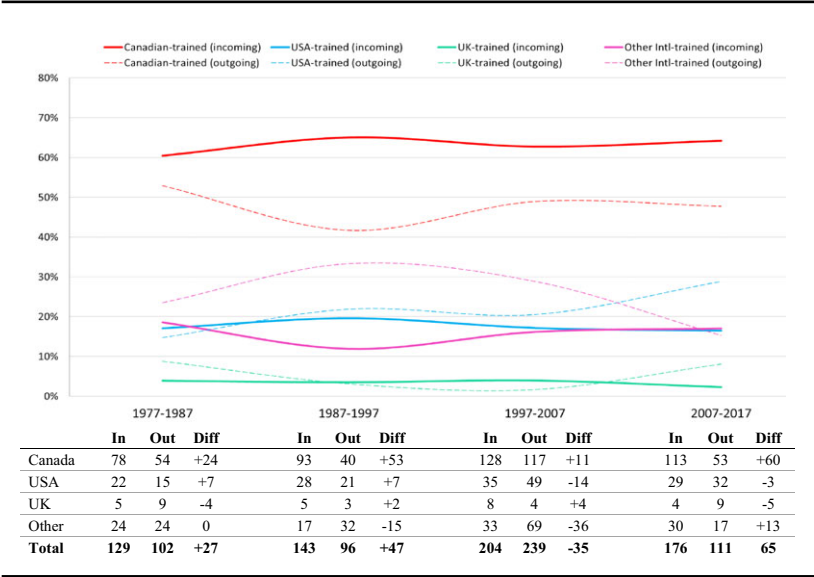
Proportions and Raw Counts of Incoming and Outgoing Faculty at Regional Universities [Color figure can be viewed at wileyonlinelibrary.com]



As for the French class, given the anglophone nature of the Canadianization Movement, this group exhibits remarkably stable hiring trends over the last four decades, holding out as the only class that maintained a majority proportion of Canadian-trained faculty since 1977 (Figure 1), roughly 70 percent of whom received their Ph.D.s from Québec's universities (results not shown). The same flatline trends are observed among new hires (Figure 5), where the roughly 60 percent Canadian-trained and 20 percent American-trained hiring trends waver very little over 40 years. Unique to the French class, they are the only universities that, despite large proportions of outgoing faculty in 1987 to 1997 (33 percent) and 1997 to 2007 (29 percent), maintained more than 10 percent of other international faculty (comprised almost entirely of French-speaking schools in France and Belgium). Furthermore, 2007 to 2017 marks the first period during which more other international trained faculty are hired than are exiting this class.

Figure 5

Proportions and Raw Counts of Incoming and Outgoing Faculty at French Universities [Color figure can be viewed at wileyonlinelibrary.com]



Ph.D. PLACEMENT

Next, going beyond Ph.D.’s national origin, Table 2 shows the 10 most successful schools in placing their Ph.D.s in Canada’s U15 universities in 1977 and 2017 (Table 2). Results reveal that these 10 schools accounted for 51.5 percent of Glocal faculty’s doctoral training in 1977 (eight American and two Canadian). For National and Regional classes, the top 10 placement schools make up only 35 and 30 percent of faculty composition in 1977, respectively, indicative of a less concentrated hiring outcomes at the time. Concerning the French class, of the 10 universities that comprise 58 percent of French faculty, 31 percent came from Montréal, Ottawa, and Laval, signifying rather strong tendencies toward inbreeding hiring practices (i.e., hiring one’s own graduates), especially for Laval and Montréal.

In 2017, the Glocal class remains heavily populated by a professoriate group trained at high-status American universities and the top three Canadian universities. This class also displays the most stable networked

Table 2

Faculty Composition by Degree Granting University in 1977 and 2017

Top 10 Universities	Country	n	%	Top 10 Universities	Country	n	%
Glocal Class 1977				Glocal Class 2017			
Harvard University	USA	43	8.6	University of Toronto	CAN	56	8.1
University of Toronto	CAN	36	7.2	Berkeley	USA	38	5.5
University of Chicago	USA	33	6.6	Harvard University	USA	33	4.8
Berkeley	USA	32	6.4	Yale University	USA	31	4.5
Yale University	USA	29	5.8	Princeton University	USA	26	3.8
Columbia University	USA	23	4.6	University of Chicago	USA	24	3.5
McGill University	CAN	18	3.6	UBC	CAN	23	3.3
University of Wisconsin	USA	15	3.0	University of Michigan	USA	21	3.0
Princeton University	USA	14	2.8	Stanford University	USA	19	2.8
Stanford University	USA	14	2.8	McGill University	CAN	15	2.2
Sub-Total		257	51.5	Sub-Total		286	41.4
Total		499	100	Total		690	100
National Class 1977				National Class 2017			
University of Toronto	CAN	42	7.1	University of Toronto	CAN	60	9.9
University of Alberta	CAN	22	3.7	McMaster University	CAN	36	5.9
McGill University	CAN	22	3.7	Western University	CAN	33	5.4
LSE	UK	21	3.5	Queen's University	CAN	27	4.4
University of Chicago	USA	20	3.4	UBC	CAN	23	3.8
Harvard University	USA	17	2.9	York University	CAN	21	3.5
University of Michigan	USA	16	2.7	University of Minnesota	USA	21	3.5
Yale University	USA	16	2.7	McGill University	CAN	19	3.1
Oxford	UK	16	2.7	University of Alberta	CAN	18	3.0
Cornell University	USA	15	2.5	Berkeley	USA	13	2.1
Sub-Total		207	35.0	Sub-Total		271	44.6
Total		592	100	Total		607	100
Regional Class 1977				Regional Class 2017			
University of Toronto	CAN	28	6.9	University of Toronto	CAN	46	10.8
University of Alberta	CAN	16	4.0	UBC	CAN	28	6.6
Berkeley	USA	12	3.0	Queen's University	CAN	22	5.2
Cornell University	USA	11	2.7	University of Calgary	CAN	19	4.5
University of Minnesota	USA	10	2.5	University of Alberta	CAN	17	4.0
LSE	UK	10	2.5	Western University	CAN	17	4.0
McGill University	CAN	9	2.2	Carleton University	CAN	15	3.5
Queen's University	CAN	9	2.2	University of Manitoba	CAN	15	3.5
University of Chicago	USA	9	2.2	McMaster University	CAN	15	3.5
University of Colorado	USA	9	2.2	University of Waterloo	CAN	14	3.3
Sub-Total		123	30.4	Sub-Total		208	48.8
Total		405	100	Total		426	100
French Class 1977				French Class 2017			
University of Montréal	CAN	51	14.5	University of Montréal	CAN	80	17.5
University of Ottawa	CAN	31	8.8	Laval University	CAN	55	12.1
Laval University	CAN	28	8.0	McGill University	CAN	42	9.2
University of Paris	Other	28	8.0	UQAM	CAN	27	5.9
Louvain	Other	22	6.3	University of Toronto	CAN	22	4.8
McGill University	CAN	10	2.8	UBC	CAN	12	2.6
LSE	UK	10	2.8	Carleton University	CAN	11	2.4
University of Toronto	CAN	9	2.6	Western University	CAN	9	2.0
University of Geneva	Other	8	2.3	Sciences Po	Other	8	1.8
University of Chicago	USA	6	1.7	University of Ottawa	CAN	7	1.5
Sub-Total		203	57.7	Sub-Total		273	59.9
Total		352	100	Total		456	100

structure, as 7 of the top 10 schools from 1977 remain in 2017. Conversely, the National, Regional, and French classes show substantial network re-composition with many new institutions represented in the top 10 in 2017. The National and Regional classes exhibit a stronger presence of Canadian-trained scholars, with eight Canadian universities accounting for close to 40 percent of the National class's faculty and 50 percent of Regional faculty coming from 10 Canadian universities. The composition of the top 10 placement universities for the French class is almost entirely Canadian in 2017, comprising close to 60 percent of their faculty—34.5 percent from Québec's French-speaking schools.

As Table 2 suggests, in-class inbreeding was not unique to the French class. The early hiring trends of Glocal schools appear to mark a disposition toward populating their professorial positions with Glocal graduates. Between 1977 and 1987, of the 60 Canadian-trained faculty hires, 23 were trained at U of T, 10 at McGill, and 6 at UBC, representing an in-class inbreeding level of 65 percent (results not shown). This pattern of high-status inbreeding among Glocal universities persists through the years, with hires consisting of 51 percent Glocal graduates in the 1987 to 1997 period, 60 percent in 1997 to 2007, and 55 percent over the most recent decade.

Finally, examining the institutional composition of American-trained faculty in 1977 (see Table 3), results show that 72 percent of the 335 American-trained faculty working at Glocal schools received their highest degree from one of 15 high-status American universities, a majority proportion that drops slightly in 2017 to 62 percent. These same 15 schools account for 49 percent of the 327 American-trained faculty working at National and 40 percent of the 230 working at Regional universities; proportions that drop to 38 and 25 percent in 2017, respectively. In terms of overall composition, these 15 high-status schools alone account for nearly half (48 percent) of all Glocal faculty's doctoral origin in 1977 and 40 percent in 2017. These figures are significantly lower for the two other classes: 27 and 15 percent for the National schools, and 23 and 6 percent for the Regional schools in 1977 and 2017, respectively.

DISCUSSION

Historical Canadianization

As far as the historical (1977–1997) wave of postwar Anglo-Canadian academic nationalism is concerned, there are two important empirical findings. First, the anglicized U15 can be divided into two groups, with Glocal on one side (displaying a fairly status quo state of affairs with a dominant proportion of American-trained faculty over time) and National and Regional classes on the other (where domestic doctoral credentials appear to

Table 3

American-Trained Faculty Composition by High-Status Degree Granting University in 1977 and 2017																
High-Status American Universities	Glocal Class						National Class						Regional Class			
	1977			2017			1977			2017			1977		2017	
	n	Percentage	n	Percentage	n	Percentage	n	Percentage	n	Percentage	n	Percentage	n	Percentage	n	Percentage
Berkeley	32	9.6	38	8.6	8	2.4	13	5.5	12	5.2	4	3.7				
Columbia University	23	6.9	10	2.3	14	4.3	2	0.8	5	2.2	1	0.9				
Cornell University	8	2.4	10	2.3	15	4.6	8	3.4	11	4.8	5	4.7				
Duke University	3	0.9	8	1.8	7	2.1	4	1.7	7	3.0	1	0.9				
Harvard University	43	12.8	33	7.4	17	5.2	6	2.5	7	3.0	3	2.8				
Johns Hopkins University	7	2.1	8	1.8	5	1.5	5	2.1	1	0.4	2	1.9				
MIT	9	2.7	9	2.0	7	2.1	1	0.4	1	0.4	1	0.9				
Northwestern University	6	1.8	15	3.4	2	0.6	7	3.0	6	2.6	0	0.0				
Princeton University	14	4.2	26	5.9	15	4.6	5	2.1	3	1.3	1	0.9				
Stanford University	14	4.2	19	4.3	7	2.1	4	1.7	6	2.6	2	1.9				
UCLA	6	1.8	12	2.7	2	0.6	8	3.4	5	2.2	2	1.9				
University of Chicago	33	9.9	24	5.4	20	6.1	11	4.6	9	3.9	1	0.9				
University of Michigan	7	2.1	21	4.7	16	4.9	8	3.4	7	3.0	0	0.0				
University of Pennsylvania	6	1.8	9	2.0	8	2.4	4	1.7	4	1.7	0	0.0				
Yale University	29	8.7	31	7.0	16	4.9	4	1.7	7	3.0	4	3.7				
Subtotal	240	72	273	62	159	49	90	38	91	40	27	25				
Total American-trained Faculty	335	100	443	100	327	100	237	100	230	100	107	100				

have taken root as dominant forms of symbolic capital). Second, our results suggest that the sheer increase in Canadian schools' doctoral production since the 1970s cannot be taken as the only force behind the overall increase of Canadian-trained professoriate; if so, we should have seen a much larger Canadianization of doctoral credentials at Glocal schools. Despite the combined effects of the Canadianization Movement, the Canadian First Policy, and the growing body of Canadian Ph.D.s on the market, the Glocal class's overall proportion of domestically trained faculty did not even reach the minimum overall proportion of 28 percent displayed by the National and Regional Classes in 1977 (Figure 1). Furthermore, the proportion of *incoming* Canadian- versus American-trained faculty diverged radically over 40 years, reaching a 50 percent difference in 2007 to 2017 (68 percent American-trained and 20 percent Canadian-trained hires; Figure 2). But why, as these trends suggest, did the Canadianization Movement not permeate the walls of the Glocal schools? Could it be, as argued by Zeitlin in his defense of U of T's controversial hiring of eight American faculty in 1974 (see Cormier 2004:148–53), simply a case of hiring Canadians with American doctoral educations? A supplementary analysis using country of bachelor degree attainment as a rough measure of inferred citizenship reveals that only 27 percent of 516 American-trained hires⁴ over the past 40 years earned their bachelor's degree from a Canadian university; evidence that Glocal schools' response to the Canadianization Movement was likely not the harmonization of U.S. doctoral credentials and Canadian citizenship. What then, may have shaped the Glocal class's timid investment in the Canadianization Movement? We consider the influence of two forces: established high-status scholarly networks and the pursuit of scholastic "quality" and "merit."

Networks and Influence

The late post-World War II Canadian academic field is described as being characterized by homophilic professoriate reproductive practices operating along established *old boys' networks* (see Axelrod 1982; Cormier 2004; Johnston 2005), where dominant fractions of foreign faculty "tended to hire individuals who were much like themselves in terms of training, outlook, approach" (Cormier 2002:13). It was also a time when headship positions held actionable decision-making power over departmental affairs (Johnston 2005; Stortz and Panayotidis 2006), as Cormier (2002) explains, it was not uncommon for departmental chairs to "basically hire new faculty from the university at which they were trained" (p. 13). While we cannot speak directly to the homophilic hiring processes that may have taken place in the 1970s, our results reveal several key differences regarding

⁴ Excluding missing bachelor degree data for 63 of the 579 total American-trained hires (11%).

the embeddedness of high-status Ph.D. networks at Glocal, National, and Regional schools. For instance, in 1977, nearly half of all Glocal faculty were trained at 15 high-status U.S. schools (72 percent of all American-trained faculty in 1977 and 62 percent in 2017), whereas high-status scholars from the same 15 U.S. schools represented roughly one-quarter of all faculty at National and Regional schools in 1977 (49 and 40 percent of all American-trained faculty), and many fewer in 2017 (38 and 25 percent, respectively). Drawing parallels with Burris's (2004) depiction of America's prestige-driven academic caste system in which high-status field position begets high status within doctoral exchange networks, Glocal's disposition for academics trained at high-status universities coupled with the fact that only one Canadian-trained faculty member was departmental chair in 1977 (results not shown), might help explain why Canadian-ness was hardly recognized as a legitimate form of symbolic capital (see also Goyder 2009). Conversely, the greater representation of Canadian-trained faculty as departmental chairs at National (six Canadian-trained heads) and Regional (five Canadian-trained heads) classes in 1977 reveals composition trends that may have been more amenable to the changes proposed by the Canadianization Movement.

Universal Principles of Scholastic Quality and Merit versus Canadian Equity

Apart from the American-trained majority of high-status scholars who also held departmental positions of influence, what else might help us understand the Glocal class's escalating tendency to hire American-trained faculty during the height of the Canadianization Movement? Part of the answer, we submit, might lie with Glocal's vision of academic "quality" and "merit" and how it fit within the context of the Canadianization Movement's commitment to "equity" in the hiring process; a tension Christine Musselin (2005) describes as the choice between "the good scholar before all" and "the good citizen" (p. 291). Glocal's increasing proportions of American-trained hires over time, most of whom come from high-status schools, suggests faculty reproduction trends leaning more heavily to the side of "the good high-status *American-trained* scholar before all," while the National and Regional classes might have seen hiring as a means to increase Canadian representation at universities. This does not necessarily mean that Glocal schools were, for instance, hiring scholars with Harvard Ph.D.s purely to accumulate symbolic capital. As Burris (2004) explains, practices of departmental prestige enhancement, though strategic, "are typically embedded within, and subjectively experienced as indistinguishable from, a normative commitment to universalistic principles" (p. 261). For Glocal schools, the universalistic principles guiding scholastic

evaluation of “merit” tended to lean toward the American model (Axelrod 1982:169).

At the time, the rhetorical defense for the hiring procedures contended that nationality and/or place of study should not interfere with established criteria of academic “quality” and “merit,”⁵ in which the best-qualified candidates would rise to the top (Barbaric and Jones 2016). For Glocal schools, the preferential hiring of Canadian-trained scholars over “better-qualified” U.S. candidates from well-positioned American institutions might have been viewed as a choice with destructive consequences (e.g., prestige deflation). This point is emphasized by U of T’s official statement to the committee on University Affairs in the late 1960s: “[W]e have to be able to recruit in the U.S. since it would damage the entire Canadian academic community if we acquired all our needed staff from other Canadian universities” (Axelrod 1982:169). To be sure, while valuation of “merit” during the application process considers many factors (Boyle et al. 2010; Musselin 2005; Richardson, McBey, and McKenna 2006), the hiring outcomes revealed in our findings (Figure 2) suggest that Glocal schools seem to be increasingly reaching the conclusion that high-status American-trained applicants possess qualities that deem them most qualified. The American universities nurturing these “exceptional” academics with qualities desirable to the Glocal class comprise the likes of Berkeley, Harvard, Yale, Princeton, University of Chicago, University of Michigan, Stanford, Northwestern, and University of Wisconsin (Tables 2 and 3). Of the Canadian-trained scholars deemed most qualified—and most suitable to fulfill the mission of the Canadianization Movement and the Canadian First Policy—our results show that the pool has been primarily filled with Glocal’s own Ph.D.s (65 percent of new Canadian hires from 1977 to 1987, 51 percent from 1987 to 1997, 60 percent from 1997 to 2007, and 55 percent from 2007 to 2017 coming from U of T, UBC, or McGill). In-class inbreeding seeming to be their risk-free solution to prestige loss.

CONTEMPORARY PERIOD: WHO ARE REPLACING THE CANADIANIZATION GENERATION?

For the contemporary period, can one talk, as Yves Gingras (2010) suggests, of the end of the Canadianization Movement from the mid- to late-1990s onward? For the Glocal and National classes, one could note an overall *trend reversal* observed in 2007 (Figure 1) and cautiously interpret the falling overall proportions of Canadian-trained faculty as a diminishing

5. See Cormier (2004:126–33) for a historical account of the degree to which Canadian academics were divided on the merit versus citizenship debate. To be sure, the debate was not a clear-cut struggle between Glocal schools and the rest of the field, as several other universities were strongly against the nationalistic undertones of the proposed policy.

Canadianization effect. Although this would be somewhat inaccurate for the Glocal class, given their strengthening disposition to hire American-trained faculty over time, marked by their largest showing of American hires in at least 40 years during the recent decade. The National class, however, is the only one to display a proper dual trend reversal of both outgoing and incoming faculty. In their case, the mass exit of Canadian-trained faculty beginning in the 1987 to 1997 decade marks a period of subtly increasing proportions of American-trained and diminishing Canadian-trained faculty hires—converging trends that are suggestive of job vacancies increasingly filled by American-trained scholars. As for Regional and French classes, the Canadianization—and Québecization—effects appear to remain strong. The proportion of outgoing Canadian-trained faculty at Regional schools bear similarities with the National class, though, in their case the vacancies continue to be filled by a majority of Canadian-trained academics. Amid the consistent incoming trends in the French class, the contemporary period is marked by a steady decrease of outgoing European-trained faculty and an increase of outgoing American- and UK-trained faculty, perhaps indicative of a re-emergent European presence.

To understand these trends, we must consider the interplay of several field-wide forces over the past 20 years: the generational faculty renewal of significant numbers of Canadian professoriate, the neutering of the Canadian First Policy,⁶ and the entrenchment of globalizing ideals and internationally driven strategic plans. Most notably, the retirement wave of Canadian-trained faculty around 1997 marks the departure of increasing proportions of faculty who were not only present during the time of the Canadianization Movement, but likely enforcers of the Canadian First Policy. It is also around this 1997 to 2007 period during which our findings show increasing proportions of outgoing Canadian-trained scholars being replaced by, for the first time in at least four decades, increases in the proportion of American-trained Ph.D.s being hired at National and Regional universities. From these trends, we contend that the entrenchment of globalist ideals as proposed by Gingras cannot explain the end of the Canadianization Movement on its own; we must also consider how the mass exit of the “Canadianization generation” may have altered departmental hiring practices. Furthermore, “international” forms of scientific capital seem to have gained currency on the Canadian academic job market, though, in the case of the Canadian field, we echo Warren’s (2014) position that “international” seems to be another name for “American”—or in the case of the French class, “European.”

⁶ The “two-tier” system was abolished in 2003, giving departments increased freedom to cross-post positions nationally and internationally during the first round of recruitment (Cormier 2004).

RHYTHMS OF THE FIELDS: BEYOND DOCTORAL ORIGINS

Since the Federal government's 2011 decision to discontinue the University and College Academic Staff System—Statistics Canada's annual census of full-time faculty in Canadian universities—there has been a shortage of information on Canada's professorial class (Charbonneau 2015) and a general lack of empirical research investigating the structure of the Canadian academic field. Our study is a first step toward establishing a better understanding of the past, present, and future rhythms of the Canadian academic job market. Grounded in the largest longitudinal sample of Canadian social scientists' doctoral credentials collected to date, our exploratory findings signal noteworthy differences among the country's top 15 research-intensive universities. The American-trained hiring trends displayed by McGill, U of T, and UBC suggest that they were in no way swept up by the Canadianization Movement and the mandates of the Canadian First Policy. Displaying great network stability, the Glocal class's disproportionate recruitment of social scientists trained at established U.S. Ph.D.-granting universities is telling of "meritocratic" standards that seem to privilege scholars imbued with the powerful prestige of high-status institutions. The Regional and French schools revealed the opposite trend with a sustained Canadianization/Québecization of their faculty's Ph.D. origin during the whole period under study. The National class trends suggest otherwise. Displaying a nationalized habitus during and beyond the Canadianization Movement's prime moment, the Americanized turn of incoming faculty replacing the Canadianization generation is suggestive of a shift privileging foreign-trained scholars over domestic. Counter to Wilkinson et al.'s (2013) conclusion that "the mythology around the preference for foreign-trained PhDs is largely unfounded, except among some of the medical/large universities" (p. 358), our findings suggest that preferences for foreign-trained hires is growing, to varying degrees, among at least eight of the country's largest and most prestigious English-speaking research-intensive universities. Whether these trends will continue over the coming years remains to be seen.

While our study has provided the first longitudinal look at the ways in which academic credentials as symbolic capital can help understand "who got the jobs" at Canadian universities over time, further multimethod initiatives on academic hiring practices are needed to identify the breadth of forces at play before, during, and after the hiring process, such as gender (Nakhaie 2008), race (Henry et al. 2017), soft skills, network positionality, and research "quality" and productivity (Goyder 2009; Musselin 2005). To this end, future research should investigate the extent to which "meritocratic" hiring practices centered on academic productivity favor certain scholarly profiles and records over others and how institutional "prestige" factors into the adjudication process. The dominant and persistent

presence of faculty trained at “prestigious” American universities, coupled with the prevalent in-class inbreeding of Glocal graduates at Canada’s top three schools, are indications that prestige may be a contributing force in the Canadian academic field. The degree to which this is true certainly warrants further investigation. Collecting data on the country’s professoriate class’s accomplishments during their doctoral tenure would reveal the extent to which the prestige of their credentials outweighs the substance of their CVs—paying close attention to high-status hires whose academic output at time of graduation consists entirely of “works in progress.” Furthermore, we need departmental level data tracking the composition and characteristics of rejected and shortlisted applicants. With anonymity and privacy concerns in mind, simple documentation of each applicant’s Ph.D. university, number of publications, research area, and/or total funding dollars received would be a step in the right direction.

Further study on the sociology of academia in Canada would also do well to move beyond U15 schools to include all research-intensive universities as well as colleges and institutes, taking steps to reveal the ways in which the stratification highlighted in this paper resonate and reverberate differently throughout the Canadian academic field. While our sampling strategy began around the emergence of the Canadianization Movement, historically inclined researchers could also go back before 1977 to document the shape of the postwar *first-wave* doctoral “Americanization” that sparked the Canadianization Movement. Our method of collecting at 10-year intervals sufficed as a first step toward understanding the processual dimensions of the Canadian academic field, but may have missed occurrences of nomadic early career academics who take up positions to progress rather than to lay down roots. Furthermore, next steps should provide more granularity and investigate differences not only between sciences, humanities, and social sciences, but within social science disciplines as well. Finally, while we have focused on “who made it” at U15 schools, we also need to investigate the extent to which Canadian-trained scholars are hired at American universities. This will allow us to better understand the balance between incoming and outgoing Ph.D. exchange networks and how Canadian “prestige” translates across international boundaries. In short, what we need is data. Greater accountability at the departmental and institutional levels will go a long way to support further research that can help the new generation of Canadian-trained scholars understand the rhythms of the game they are playing.

References

- Axelrod, P. 1982. *Scholars and Dollars: Politics, Economics, and the Universities of Ontario, 1945–1980*. Toronto, ON: University of Toronto Press.

- Barbaric, D. and G.A. Jones. 2016. "International Faculty in Canada: Recruitment and Transition Process." Pp. 51–75 in *International Faculty in Higher Education: Comparative Perspective on Recruitment, Integration and Impact*, edited by M. Yudkevich, P.G. Altbach L.E. Rumbley. Abingdon, Oxfordshire: Routledge.
- Bell, S. and J. Gordon. 1999. "Scholarship—The New Dimension to Equity Issues for Academic Women." *Women's Studies International Forum* 22(6):645–58.
- Blackwell, T. 2017. "Brain Drain: Brexit and Trump Have Top-Notch Academics Fleeing—and Canadian Schools Stand to Gain." Retrieved April 4, 2017 (<http://news.nationalpost.com/news/canada/brain-drain-brexit-and-trump-have-top-notch-academics-fleeing-and-canadian-schools-stand-to-gain>).
- Boyle, J.G., J.J. Furedy, D.L. Neumann and H. Rae. 2010. "Balance Between Merit and Equity in Academic Hiring Decisions: Judgemental Content Analysis Applied to the Phraseology of Australian Tenure-Stream Advertisements in Comparison with Canadian Advertisements." *Australian Universities' Review* 52(2):49–55.
- Brown, D.G. 1967. *The Mobile Professors*. Washington, DC: American Council on Education.
- Brym, R. 2002. "An Introduction to the Upper Thirteen." *American Sociologist* 33(1):5–11.
- Burris, V. 2004. "The Academic Caste System: Prestige Hierarchies in PhD Exchange Networks." *American Sociological Review* 69(2):239–64.
- Charbonneau, L. 2015. "Filling in the Postsecondary Education Data Gaps." Retrieved January 5, 2017 (<https://www.universityaffairs.ca/opinion/margin-notes/filling-in-the-postsecondary-education-data-gaps/>).
- Chiswick, B.R., N. Larsen and P.J. Pieper. 2010. "The Production of PhDs in the United States and Canada." Institute for the Study of Labor (IZA), Bonn, Germany. Unpublished manuscript.
- Cormier, J. 2002. "Nationalism, Activism, and the Canadian Sociology and Anthropology Community, 1967–1985." *American Sociologist* 33(1):12–26.
- Cormier, J. 2004. *The Canadianization Movement: Emergence, Survival, and Success*. Toronto, ON: University of Toronto Press.
- Dandurand, P. 1989. "Un corps professionnel renouvelé: Les professeurs des universités Québécoises francophones entre 1959 et 1976." *Canadian Journal of Higher Education* 19(1):63–85.
- Davies, S. and F. Hammack. 2005. "Channeling Competition in Higher Education: Comparing Canada and the US." *Journal of Higher Education* 76:89–106.
- Desjardins, L. 2012. *Profile and Labour Market Outcomes for Doctoral Graduates from Ontario Universities*. Toronto, ON: Higher Education Quality Council of Ontario.
- Field, C.C., G.A. Jones, G. Karram Stephenson and A. Khoyetsyan. 2014. *The "Other" University Teachers: Non-Full-Time Instructors at Ontario Universities*. Toronto, ON: Higher Education Quality Council of Ontario.
- Fischer, D. 1991. *The Social Sciences in Canada: 50 Years of National Activity by the Social Science Federation of Canada*. Waterloo, ON: Wilfred Laurier University Press.
- Gingras, Y. 2010. "The End of the Canadianization Movement: A Globalization By-Product?" Retrieved January 5, 2017 (<http://www.universityaffairs.ca/opinion/in-my-opinion/end-of-the-canadianization-movement/>).
- Goyder, J. 2009. "A Behavioural Approach to Ranking Academic Departments by Prestige: The Case of Sociology in English-Speaking Canada." *Bulletin de Méthodologie Sociologique* 101:28–41.

- Groarke, L. and W. Fenske. 2009. "PhD: To What End?" Retrieved January 4, 2017 (<http://www.universityaffairs.ca/features/feature-article/phd-to-what-end/>).
- Guppy, N., E. Grabb, and C. Mollica. 2013. "The Canada Foundation for Innovation, Sociology of Knowledge, and the Re-Engineering of the University." *Canadian Public Policy* 39 (1):1–19.
- Hagan, J. 2001. *Northern Passage. American Vietnam Resisters in Canada*. Boston, MA: Harvard University Press.
- Henry, F., E. Dua, A. Kobayashi, C. James, P. Li, H. Ramos and M.S. Smith. 2017. "Race, Racialization and Indigeneity in Canadian Universities." *Race, Ethnicity and Education* 20(3):1–15.
- Johnston, J. 2005. "The 'Second Shift' of Canadian Sociology: Setting Sociological Standards in a Global Era." *Canadian Journal of Sociology* 30(4):513–27.
- Looker, D.E. 2015. "Appendix to 41st Statistical Report." Canadian Association for Graduate Studies. Unpublished manuscript.
- Mathews, R. 2014. *The Canadianization Struggle: A History*. Vancouver, BC: Northland Publications.
- Mathews, R. and J.A. Steele. 1969. *The Struggle for Canadian Universities: A Dossier*. Toronto, ON: New Press.
- McLaughlin, N. 2005. "Canada's Impossible Science: Historical and Institutional Origins of the Coming Crisis in Anglo Canadian Sociology." *Canadian Journal of Sociology* 30(1):1–40.
- McLaughlin, N. 2006. "Whither the Future of Canadian Sociology? Thoughts on Moving Forward." *Canadian Journal of Sociology* 31(1):107–30.
- Musselin, C. 2005. *Le Marche des Universitaires: France, Allemagne, États-Unis*. Paris: Sciences Po les Presses.
- Nakhaie, M.R. 2008. "Universalism, Ascription and Academic Rank: Canadian Professors, 1987–2000." *Canadian Review of Sociology and Anthropology* 44(3):362–86.
- Rajagopal, I. 2002. *Hidden Academics: Contract Faculty in Canadian Universities*. Toronto, ON: University of Toronto Press.
- Research Infosource. 2017. "Canada's Top Research Universities." Retrieved March 15, 2017 (<https://www.researchinfosource.com/>).
- Richardson, J., K. McBey and S. McKenna. 2006. "International Faculty in Canada: An Exploratory Study." SRI Working Paper Series No. 2006 D-22.
- Robertson, R. 1995. "Glocalization: Time-Space and Homogeneity-Heterogeneity." Pp. 25–44 in *Global Modernities*, edited by M. Featherstone, S. Lash and R. Robertson. London: Sage Publication.
- Siler, K.S. and N. McLaughlin. 2008. "The Canada Research Chairs program and Social Science Reward Structures." *Canadian Review of Sociology* 45(1):93–119.
- Stack, M. 2016. *Global University Rankings and the Mediatization of Higher Education*. London: Springer.
- Statistics Canada. 1973. "Education in Canada, Statistics Canada Catalogue No. 81–229." Retrieved June 2, 2017 (https://www.statcan.gc.ca/access_acces/archive.action?l=eng&loc=W504_512-eng.csv).
- Stortz, P. and E.L. Panayotidis. 2006. *Historical Identities: The Professoriate in Canada*. Toronto, ON: University of Toronto Press.
- Symons, T.H.B. 1975. *To Know Ourselves*. The Report of the Commission on Canadian Studies, Vols. I and II. Ottawa, ON: Association of Universities and Colleges of Canada.

- Teichler, U., A. Arimoto and W.K. Cummings. 2013. *The Changing Academic Profession: Major Findings of a Comparative Survey*. New York: Springer.
- Tremblay, R., S. Hardwick and J. O'Neill. 2014. "Academic Migration at the Canada–U.S. Border." *American Review of Canadian Studies* 44(1):118–34.
- Warren, J. 2014. "The End of National Sociological Traditions? The Fates of Sociology in English Canada and French Québec in a Globalized Field of Science." *International Journal of Canadian Studies* 50:87–108.
- Wilkinson, L., J. Bramadat, R. Dolynchuk and Z.T. St. Aubin 2013. "Are Canadian-Trained PhDs Disadvantaged in the Academic Labor Market?" *Canadian Review of Sociology* 50(3):357–70.
- Williams, G. 2005. "Doctoral Education in Canada, 1900–2005." Ottawa, Canadian Association for Graduate Studies. Unpublished manuscript.
- Wood, M. 2017. "What is the Value of a Canadian PhD to Universities in Canada?" Retrieved April 5, 2017 (<http://www.universityaffairs.ca/career-advice/career-advice-article/a-study-tenure-track-faculty-in-history-and-english-canada/>).
- Worswick, C. 2005. "Mandatory Retirement Rules and the Retirement Decisions of University Professors in Canada." *Labour Economics* 17(6):1022–29.